

# Factorial-weighted Stirling products (OEIS A192563 and A192561)

Diagonal poly-Cauchy numbers at all fixed orders, and matched cycles:  
exact certificates, all orders and inverses

Two research reports of one external research session  
(bundle Reports 154 and 155), merged into one ProveIt report

Both Parts: 3 October 2026   merged: 6 October 2026

## Abstract

This report prints two manuscripts as Parts I and II. They treat two members of one OEIS family entered by Emanuele Munarini on 4 July 2011, sums over  $k$  of  $k!$  times a product of two Stirling numbers. Part I treats A192563,  $a_n = \sum_k k! \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} \{n+1\}_k = \sum_m \begin{bmatrix} n \\ m \end{bmatrix} (m+1)^n$ , the diagonal of the poly-Cauchy array A344639 of Bényi and Ramírez;  $a_n/n!$  is the  $n$ th moment of one plus the cycle count of a uniform random permutation. It proves an asymptotic expansion of  $a_n$  to every fixed order in  $r/n$ , where  $r = \log n - \log \log n - \log \log \log n + o(1)$  is the exact saddle of  $e^z \Gamma(n + e^z) / \Gamma(e^z)$  and the coefficients are explicit Gaussian contraction polynomials in exact cumulants; the same expansion for the entries  $A_{n,k}$  of A344639, uniformly for  $k/n$  in a compact subset of  $(0, \infty)$ ; a smooth inverse with floor and ceiling bounds for both threshold indices, correct at equality cases; and a nested-logarithm inverse. Part II treats A192561,  $b_n = \sum_k k! \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}^2$ , as the Gaussian Fock norm of  $\sum_k \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} z^k$ . It proves an exact positive expansion with an explicit tail at every  $n \geq 1$ , a rational-centre certificate that isolates  $b_{10} = 1273270076895432$ , the same certificate for finite products with negative real zeros, an expansion to every fixed order in  $1/w$ ,  $w = W_0(en)$ , with rational coefficients, and a nested-Lambert inverse with two-ceiling threshold bounds. The two manuscripts use different methods and do not cite each other. The write adds, for the third member A192562, which neither treats, the identity  $A192562(n) = a_n - nA_{n-1,n}$ , and states what the two methods suggest for its asymptotics. Effective constants and onsets, growing orders, the  $1/n$ -scale correction for A192561 and the asymptotics of A192562 remain open. The report is unrefereed, and nothing in it is formalized.

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## Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 3988bf5c4 (batch 107, cluster 107-STIRL). They treat two members of one OEIS family, sums over  $k$  of  $k!$  times a product of two Stirling numbers, entered by Emanuele Munarini on 4 July 2011 (quoted below):

| Part | Bundle report | Delivered title                                                                                                            | Labels         |
|------|---------------|----------------------------------------------------------------------------------------------------------------------------|----------------|
| I    | 154 (base)    | Diagonal poly Cauchy permutations at all fixed orders: Exact saddle expansion and inverse thresholds for OEIS A192563      | <b>sfp:pc:</b> |
| II   | 155           | Exact positive certificates and all orders for matched cycles: Factorial weighted squared Stirling numbers in OEIS A192561 | <b>sfp:mc:</b> |

*Arrangement.* The base of the merge is Report 154; its `Report154.tex` was staged as this `article.tex`, and it is printed first, as Part I. It treats the whole two-parameter array A344639 of which its sequence A192563 is the diagonal. Report 155 is printed second, as Part II. Neither Part uses, cites or restates the other, so the order is that of the bundle numbers and of the generality of the object, not a dependency. Why two manuscripts with different methods make one report is explained below (“One family, two methods”).

*Numbering.* Sections are numbered continuously, and statements and equations within sections. Part I keeps Report 154’s Sections 1–11 and all its numbers. Report 155’s Section  $k$  is Section  $k + 11$  here (Sections 12–22), and its statement or equation  $k.j$  is  $(k + 11).j$ : its Lemma 3.1 and Theorems 3.2 and 3.3 are Lemma 14.1 and Theorems 14.2 and 14.3, its Theorem 4.1 is Theorem 15.1, its Proposition 6.1 is Proposition 17.1, its Theorems 7.1 and 7.2 are Theorems 18.1 and 18.2, and its Theorem 8.1 is Theorem 19.1. Its Table 1 is Table 2. Section 23, which closes the report, was added in the write; no delivered number moved.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date (6 October 2026). They connect the two Parts, record what the write checked, and say how each inverse statement relates to the repository’s transseries volume. No delivered statement was changed and no delivered symbol was renamed. Each Part ends with its own section of open questions (Sections 11 and 22); Section 23 gathers them for the whole report under Vladimir’s standing rule of 4 October 2026, with their sources, sketches and what is missing, and it treats the third member of the family, A192562, which neither manuscript treats.

## What is proved, and where

Here  $[n_m]$  and  $\{n_m\}$  are the unsigned Stirling numbers of the first kind and the Stirling numbers of the second kind;  $a_n$  is A192563 and  $A_{n,k}$  the array A344639 (Part I);  $b_n$  is A192561 and  $P_n(z) = \prod_{j=1}^n (1 + z/j)$  (Part II).

| Statement                                                                                                                                                                                                                                                                                                                                                    | Where                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| $a_n = \sum_m \binom{n}{m} (m+1)^n = \sum_k k! \binom{n+1}{k+1} \{ \frac{n+1}{k+1} \}$ , Munarini's form, for every $n$ ; $a_n/n! = \mathbb{E}(C_n + 1)^n$ ; the poly-Cauchy interpretation (prior art, credited)                                                                                                                                            | Part I, (1.1), (1.2), (2.1)                                |
| The exact saddle $r$ : existence, uniqueness, smoothness; $r = \log N - \log \log N - \log \log \log N + o(1)$ ; $\kappa_j \sim N r^{j-1}$                                                                                                                                                                                                                   | Part I, Lemmas 3.1–3.3                                     |
| $a_n$ to every fixed order in $r/n$ , with exact cumulants and explicit Gaussian contraction polynomials $E_q$ ; the whole complementary circle by a positive exponential sum                                                                                                                                                                                | Part I, Theorem 1.1, Lemma 4.2, (5.1)                      |
| The same expansion for $A_{n,k}$ , uniformly for $\alpha \leq k/n \leq \beta$                                                                                                                                                                                                                                                                                | Part I, Theorem 1.2                                        |
| Smooth model $Y_R$ eventually increasing;                                                                                                                                                                                                                                                                                                                    | Part I, Theorem 1.3, Sections 7–9, (8.1)                   |
| $\lceil \nu_R - \varepsilon_R \rceil + 1 \leq T_{\geq} \leq T_{>} \leq \lceil \nu_R + \varepsilon_R \rceil$ (asymptotic, no effective onset); nested-logarithm inverse with relative remainder $O(m^2/\ell^2)$ ; $a_{n+1} > a_n$                                                                                                                             |                                                            |
| $b_n/(n!)^2 = \ P_n\ _{\mathcal{F}}^2$ (classical Fock identity); exact positive expansion with explicit tail for every $n \geq 1$ and $K \geq 2$ ; $S_2 \leq 1/\varphi$                                                                                                                                                                                     | Part II, (13.4), Lemma 14.1, Theorem 14.2                  |
| The same certificate for finite products with negative real zeros (priority not established)                                                                                                                                                                                                                                                                 | Part II, Theorem 14.3                                      |
| Rational-centre certificate; $b_{10} = 1273270076895432$ isolated in an interval of width $< 2.3 \cdot 10^{-18}$                                                                                                                                                                                                                                             | Part II, Theorem 15.1, (15.10)                             |
| Exact-centre asymptotics and the two-dimensional Gaussian approximation                                                                                                                                                                                                                                                                                      | Part II, Section 16                                        |
| Reduction to a universal Gamma norm with relative error $O(w^2/n)$ ; $b_n$ to every fixed order in $1/w$ , $w = W_0(en)$ , with rational $A_0, \dots, A_{10}$                                                                                                                                                                                                | Part II, Proposition 17.1, Theorems 18.1 and 18.2, Table 2 |
| $b_{n+1} > b_n$ ; leading inverse $u = L/(2W_0(L/(2e)))$ ; nested-Lambert centre $v_R$ with a two-ceiling bracket (asymptotic, no effective onset); a finite certificate protocol A192562:                                                                                                                                                                   | Part II, Section 19, Theorem 19.1                          |
| $A192562(n) = a_n - nA_{n-1,n} = \sum_m \binom{n-1}{m} \{(m+2)^n - (m+1)^n\}$ for $n \geq 1$                                                                                                                                                                                                                                                                 | [write] Proposition 23.1                                   |
| <i>Open</i>                                                                                                                                                                                                                                                                                                                                                  |                                                            |
| Effective constants and onsets; the regimes $k/n \rightarrow 0$ and $k/n \rightarrow \infty$ of A344639; limit laws under the moment weights; growing orders, optimal truncation and Borel summability; the $1/n$ -scale correction for A192561; other root classes and weights; the asymptotics of A192562; full-text priority comparisons. See Section 23. |                                                            |

The finite computations of the two companions (exact integers and rationals) check identities, coefficients, finite values and one interval certificate; no asymptotic statement rests on them. Part II's rational interval for  $b_{10}$  is a proof of that one value, from Theorem 15.1, not a numerical observation.

## The OEIS family

The three entries, read by the write on 6 October 2026 (A192561 revision #10 of June 25, 2024; A192562 revision #4 of March 30, 2012; A192563 revision #13 of May 26, 2021), each with the author line `_Emanuele Munarini_, Jul 04 2011`. Their names, verbatim:

- **A192561:** `a(n) = Sum_{k=0..n} Stirling1(n+1,k+1)^2*k!.`

- **A192562:**  $a(n) = \text{sum}(\text{abs}(\text{stirling1}(n,k)) * \text{stirling2}(n+1,k+1) * k!, k=0..n).$
- **A192563:**  $a(n) = \text{Sum}_{\{k=0..n\}} \text{abs}(\text{Stirling1}(n+1,k+1)) * \text{Stirling2}(n+1,k+1) * k!.$   
with the cross-reference **Diagonal** of the array **A344639**.

So all three are  $\sum_k k! X_k Y_k$  over a pair of Stirling rows: two first-kind rows  $\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}$  (A192561; the square makes the sign of **Stirling1** irrelevant), a first-kind and a second-kind row, both shifted (A192563), or both with the first row unshifted,  $\begin{bmatrix} n \\ k \end{bmatrix} \begin{Bmatrix} n+1 \\ k+1 \end{Bmatrix}$  (A192562). The pattern is one, but A192562's index shift differs from the other two. Part II treats A192561, Part I treats A192563, neither treats A192562. The listed terms (17 in each entry) agree with the write's exact recomputation from these definitions, and A192563's b-file ( $0 \leq n \leq 270$ , Stefano Spezia) agrees with Part I's (1.1) at every index. None of the three entries states an asymptotic formula.

*Munarini's form of A192563.* Part I defines  $a_n = \sum_m \begin{bmatrix} n \\ m \end{bmatrix} (m+1)^n$  (1.1). The diagonal  $k = n$  of its identity (2.1),  $A_{n,k} = \sum_j j! \begin{bmatrix} n+1 \\ j+1 \end{bmatrix} \begin{Bmatrix} k+1 \\ j+1 \end{Bmatrix}$ , is word for word the entry's definition, and that identity holds for all  $n, k \geq 0$ : it is Theorem 2.6 of Bényi–Ramírez [3], and Part I's Subsection 2.1 reproves it in a few lines, a complete proof (the one step it calls standard is proved in a dated note there). So the two forms agree for every  $n$ , not only for the  $n \leq 11$  checked at intake, the  $n, k \leq 64$  of the companion and the  $n, k < 30$  of the write.

**A344639.** The entry, read on 6 October 2026 (revision #35 of April 15, 2025; Stefano Spezia, May 25, 2021), reads, verbatim:

- name: Array read by ascending antidiagonals:  $A(n, k)$  is the number of  $(n, k)$ -poly-Cauchy permutations.
- first formula (beginning):  $A(n, k) = \text{Sum}_{\{m=0..n\}} \text{abs}(S1(n, m)) * (m + 1)^k$
- cross-reference: Main diagonal gives A192563.

So  $A(n, k)$  is Part I's  $A_{n,k}$  (1.1) with the same indices; the write recomputed the 55 listed entries. The entry cites the arXiv preprint of Bényi–Ramírez by its own theorem numbers; Part I cites the final version. What Part I proves about the array:

- the identities it uses, all credited as prior art: the moment form (1.2), the Stirling product (2.1) (the entry's formula “Theorem 14”), the coefficient form (2.4) and the double generating function (2.5) (consistent with the entry's column generating functions of April 2025);
- Theorem 1.2: for every fixed order  $R$  and every compact  $[\alpha, \beta] \subset (0, \infty)$ , an asymptotic expansion of  $A(n, k)$  in powers of  $r/k$  with exact cumulants, uniformly for  $\alpha \leq k/n \leq \beta$ , together with the saddle scale (1.12); Theorem 1.1 is its main diagonal.

It proves nothing for fixed  $n$  (rows,  $k/n \rightarrow \infty$ ) or fixed  $k$  (columns,  $k/n \rightarrow 0$ ), which it states explicitly and lists as a question (Section 23, item 3), and nothing about the antidiagonal sums **A344640**. The entry contains no asymptotic formula. *Nothing was submitted to the OEIS.*

## One family, two methods

The two manuscripts were written in one session, carry the same date and the same release tooling, and do not cite each other. They are printed as one report because they treat two members of one

OEIS family, and because one bilinear form and one polynomial put both in the same frame. In Part II's notation (18.3),  $\langle f, g \rangle_{\mathcal{F}} = \sum_k k! [t^k] f [t^k] g$ , and with

$$n!P_n(z) = \sum_k \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} z^k \quad (13.4), \quad \mathfrak{T}_n(z) = \sum_k \left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} z^k, \\ z^{\bar{n}} = z(z+1) \cdots (z+n-1) = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} z^k,$$

the three definitions read

$$A192561(n) = \langle n!P_n, n!P_n \rangle_{\mathcal{F}}, \quad A192563(n) = \langle n!P_n, \mathfrak{T}_n \rangle_{\mathcal{F}}, \quad A192562(n) = \langle z^{\bar{n}}, \mathfrak{T}_n \rangle_{\mathcal{F}}$$

(the notation  $z^{\bar{n}}$  and  $\mathfrak{T}_n$  is the write's). Part II's sequence is a norm, and Part II exploits exactly that: a translated Gaussian norm is a sum of nonnegative terms, which gives an exact expansion with an explicit tail. Part I's sequence is a pairing of two different polynomials, and Part I does not use the Fock form at all. It works with the summed form  $A_{n,k} = \sum_m \begin{bmatrix} n \\ m \end{bmatrix} (m+1)^k$ , equal to the pairing by (2.1), and extracts a coefficient,

$$a_n = n! [z^n] F_n(z), \quad F_n(z) = e^z \prod_{q=0}^{n-1} (e^z + q) = n! e^z P_n(e^z - 1),$$

the last form being Part II's polynomial after the substitution  $z \mapsto e^z - 1$  (both products are  $\prod_{j=1}^n (j + e^z - 1)$ ). Positivity enters Part I differently:  $F_n$  is a positive exponential sum (2.2), which controls the whole circle (Lemma 4.2).

The scales differ accordingly. Part I's saddle is the exact root  $r = \log n - \log \log n - \log \log \log n + o(1)$  in the variable of  $e^z$ , and its expansion is in powers of  $r/n$  with coefficients depending on  $n$  through exact cumulants. Part II's saddle  $a \sim \log n$  and its anchor  $w - 1$ ,  $w = W_0(en)$ , give an expansion in  $1/w$  with rational coefficients, after a reduction whose error  $O(w^2/n)$  is beyond all orders in  $1/w$ . The leading growths share one factorial core:  $\log b_n = 2n \log n - 2n + O((\log n)^2)$  (19.2), while Lemma 9.1 and (8.2) give  $\log a_n = 2n \log n - n \log \log n - 2n + O(n \log \log n / \log n)$ ; so  $\log(a_n/b_n) = -n \log \log n + O(n \log \log n / \log n)$ . Both Parts prove every fixed order and integer-safe inverse brackets, both leave constants and onsets non-effective, and both state their priority searches as bounded. No lemma is shared, and neither manuscript applies its method to the other's sequence. The third member, A192562, is the subject of Section 23, item 1, where the write proves an exact identity for it.

## The inverses and the transseries volume

How each inverse statement relates to the transseries volume [19] (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`), statement by statement. No inverse here is an instance of the volume's Lambert core `p0:thm:lambert-core`.

- Part II, (19.3): the equation  $2u(\log u - 1) = L$  is the factorial core  $X(\kappa \log X + d) = L$  of `p0:prop:factorial-core` with  $\kappa = 2$ ,  $d = -2$ . That proposition gives  $v = W_0((L/2)e^{-1}) = W_0(L/(2e))$  and  $X = e^{v+1}$ , and  $e^v = L/(2ev)$  gives  $X = L/(2v)$ , which is (19.3); the side condition  $2 \log X - 2 > 0$  selects the solution  $u > e$ . An *instance*.

- Part II, Theorem 19.1 and (19.8): the centre  $v_R = u - J_R(u)/(2D)$  is a one-step correction of the core solution, proved from the smooth model  $F_R$  of (19.4), whose terms in  $w(x) = W_0(ex)$  lie outside the volume's exponential-power model `p0:thm:model1`. The two-ceiling brackets have the form of `p0:thm:staircase` (2), but that theorem presupposes an admissible interpolation of the sequence itself, and  $F_R$  is a model with an error  $\pm D_R w^{-R}$ , not an interpolation of  $(b_n)$ . Not an instance; proved directly; analogous to the separation condition (2).
- Part I, Theorem 1.3, bracket (1.17): likewise rounded from the smooth model  $Y_R$ , which is not an interpolation of  $(a_n)$ ; not an instance of `p0:thm:staircase`, analogous to its condition (2). The lower edge  $\lfloor \nu_R - \varepsilon_R \rfloor + 1$  and the strict threshold  $T_>$  are Part I's own refinements (Section 8).
- Part I, nested inverse (1.18): the model  $\log Y_R(N) = 2N \log N - N \log \log N - 2N + O(N \log \log N / \log N)$  (9.1) contains the term  $-N \log \log N$ , outside the factorial core  $X(2 \log X - 2)$ ; *not* an instance of `p0:prop:factorial-core`. After the change of variables  $\xi = \log \log X$ ,  $Z = \log N$  it reads  $\xi = Z + \log Z + \log 2 + B(Z)$  with  $B = b_1(\log Z)/Z + O((\log Z)^2/Z^2)$ ,  $b_1(\lambda) = -\lambda/2 - 1$ ; this is the monomial-logarithmic datum  $(\rho, \mu, \beta) = (1, 1, \log 2)$  of `plt:def:lw-monomial-log-datum`, but only its first coefficient  $b_1$  is determined. The write checked that the first two coefficients which `plt:thm:lw-template` produces from  $(1, 1, \log 2, b_1)$ , namely  $q_0 = -\log X'$  and  $q_1 = \frac{3}{2} \log X' + 1$  in the template's variable  $X' = \xi - \log 2$ , give exactly (1.18). The template is a formal statement about a complete series  $B$ ; Part I's remainder  $O(m^2/\ell^2)$  is its own mean-value argument (Section 9). An analogue, with first-order agreement; not an instance.

Neither manuscript claims novelty for the inversion mechanics; both call the underlying tools classical.

## Notation across the two Parts

Common to both Parts, with the same meaning:  $n$  the index; the unsigned Stirling numbers of the first kind (Part I writes `\sone`, Part II `\stir`; they print the same symbol); natural logarithms;  $\psi$  and its derivatives;  $[z^k]$ ; the DLMF as the source of the Gamma-function inputs. Part II alone uses  $W_0$ ,  $\zeta(\cdot, w)$ , the golden ratio  $\varphi$  and the Fock norm  $\|\cdot\|_{\mathcal{F}}^2$ ; Part I alone uses the second-kind numbers  $\{n\}_k$  (the front matter and Section 23 use both). The symbols below change meaning between the Parts or within one; each Part's text fixes its meaning there, and each Part opens with a short table of its own readings. No symbol was renamed.

| Symbol | Meanings                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a$    | Part I: $a_n$ the sequence A192563. Part II: $a = a_n > 0$ the saddle point of (12.2), $a \sim \log n$ . <i>Tempting false reading</i> : Part II's $a_n$ is a real number near $\log n$ , not A192563.                                                                                                                                                                                                                                                        |
| $b, B$ | Part I: $b = \kappa_2$ the second cumulant at the saddle; $B_j(p)$ the Bernoulli cumulant polynomials (3.13). Part II: $b_n$ the sequence A192561; $b \geq 0$ a translation centre (Sections 13 and 15); $b_* = w - 1$ ; $B_n = b_n/(n!)^2$ ; $B_{2j}$ Bernoulli numbers; $B_{m+1}(t)$ Bernoulli polynomials with $B_1(t) = t - \frac{1}{2}$ (18.1). <i>False reading</i> : Part II's $b$ is not a variance, and its $B_{m+1}(t)$ are not Part I's $B_j(p)$ . |
| $A$    | Part I: $A_{n,k}$ the array A344639. Part II: $A_r$ the coefficients of Theorem 18.2 and Table 2; $A$ planar Lebesgue measure in (13.2).                                                                                                                                                                                                                                                                                                                      |

| Symbol                        | Meanings                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $r, t$                        | Part I: $r = r(N)$ the exact saddle (1.4); $t = e^r$ . Part II: $r$ an index ( $A_r, S_r, H_r, p_r$ ) and a radius in the proof of (13.3); $t$ the complex translation variable. <i>False reading:</i> Part II's $A_r/w^r$ has nothing to do with Part I's saddle.                                                                                                                                          |
| $E$                           | Part I: $E_q$ the contraction polynomials (1.6), $E_1 \sim -r/(12N)$ ; $\mathbb{E}$ expectation. Part II: $E(x)$ , $E_r$ the Stirling-series coefficients (18.8), $E_1 = -1/6$ ; $E_M^\pm(x)$ exponential enclosures (15.5); $E = qd^2/(s(1-q))$ .                                                                                                                                                          |
| $C, c$                        | Part I: $C_n$ the cycle count of a uniform permutation; $C_R, C_j$ constants; $c = \frac{3}{2}m + 1 + \log 2$ (9.4); $c_j$ bounded coefficients. Part II: $C_{n,K}$ and $C_n$ the truncated and full normalized norm (14.4); $C$ a finite rational sum; $C_R$ constants; $c_k, c_k(b)$ Taylor coefficients.                                                                                                 |
| $L$                           | Part I: $L_N(z)$ of (1.3); $L = \log X$ . Part II: $L(x) = -\log(1-x)$ ; $L = \log T$ ; $\mathcal{L}$ a lower endpoint (15.8).                                                                                                                                                                                                                                                                              |
| $H, h$                        | Part I: $H(s, u)$ , $H_d(u)$ (4.9), (5.2); $h = r/N$ ; $h_N(u)$ (3.2). Part II: $H_b(t)$ (13.5); $H_r$ the norm coefficients (18.3); $\mathcal{H}(w)$ (17.2); $H_n$ a harmonic number (proof of Lemma 14.1); $h_w(t)$ , $h_{n,w}(t)$ , $h_k(w)$ ; $h$ an upper bisection endpoint.                                                                                                                          |
| $P, Q$                        | Part I: $P_R(N) = \sum_{q < R} E_q$ (1.14); $Q(u)$ (4.8), $Q_w(u)$ . Part II: $P_n(z) = \prod_{j \leq n} (1 + z/j)$ , $P(z)$ in Theorem 14.3; $p_r(t)$ (18.2); $Q_R(w) = \sum_{r < R} A_r w^{-r}$ .                                                                                                                                                                                                         |
| $F, Y, \nu, v$                | Part I: $F_n(z)$ the coefficient function (2.3); $Y_R(N)$ the smooth model (1.14); $\nu_R$ its inverse. Part II: $F_R(x)$ the smooth <i>logarithmic</i> model (19.4); $v_R$ the inverse centre (12.9); $y_R$ the smooth root. <i>False reading:</i> $F_R(n)$ approximates $\log b_n$ , while $Y_R(n)$ approximates $a_n$ itself.                                                                            |
| $N, T, X$                     | Part I: $N$ a real size parameter; $X$ the target; $T_{\geq}(X)$ , $T_{>}(X)$ the nonstrict and strict thresholds. Part II: $N(T) = \min\{n : b_n \geq T\}$ , the nonstrict threshold (Part II has no strict one); $T$ the target; $T_R$ an onset.                                                                                                                                                          |
| $q, K, D$                     | Part I: $q$ the index of $E_q$ and of $p_q$ (3.12); $q_N(t)$ (3.1). Part II: $q \in (S_2, 1)$ the scale of the tail bound; $K$ the truncation; $D = \log u$ , $D_R$ a constant.                                                                                                                                                                                                                             |
| $\lambda, \eta, \rho, \delta$ | Part I: $\lambda_j = \kappa_j/b^{j/2}$ (4.6), $\lambda = k/n$ (Section 6); $\eta = \sqrt{r/n}$ ; $\rho$ a disk radius; $\delta = n^{-2/5}$ . Part II: $\lambda_j > 0$ the roots of Theorem 14.3; $\eta_m$ exponent coefficients (18.12).                                                                                                                                                                    |
| $u, s, w, x$                  | Part I: $u = \sqrt{b}\theta$ and integration variables; $s$ an auxiliary variable (4.9); $w$ the index of $Q_w$ ; $x = N/t$ , $x_0$ (9.4), $x_j$ indeterminates. Part II: $u = L/(2W_0(L/(2e)))$ ; $w = W_0(en)$ , $w_u = W_0(eu)$ ; $s = S_2(b)$ ; $x$ a generating-function variable, $x_j = (j+b)^{-1}$ , $x_{\pm}$ model roots.                                                                         |
| $m, J, M, U, R$               | Part I: $m = \log \ell$ ; $m_j$ multiplicities; $J$ a total degree; $M_R$ an integer, $M_n$ the diagnostic normalizer (10.1); $U$ a cutoff; $\mathcal{R}$ a remainder. Part II: $m$ the degree in Theorem 14.3 and an integer in (15.9); $J$ a number of bisection steps and $J_R(u)$ (19.9); $M$ a Taylor order; $\mathcal{U}$ an upper endpoint; $R_{n,w}(t)$ (17.7). $R$ is a fixed order in both Parts. |
| $\varepsilon, g, G, d, S, Z$  | Part I: $\varepsilon_R$ (1.16); $G(u)$ (3.5). Part II: $\varepsilon_{n,K}$ , $\varepsilon_{b,K}$ tails; $g(a)$ , $g_m(t)$ (18.1); $d, d_w, d_{n,w}$ linear coefficients; $S_r, S_r(b)$ moments; $Z = (n!)^2 P_n(b)^2$ .                                                                                                                                                                                     |

## Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243. Their author lines read “Report 154” and “Report 155”; neither names an author, a tool or an addressee, and each sets an empty PDF author field. Neither names a ProveIt commit or path, so neither carries a pin. They arrived in commit 60f54ea06 and were placed by 3988bf5c4 (batch 107, cluster 107-STIRL).

- **Part I**, bundle Report 154, archive `Diagonal_Poly_Cauchy_All_Orders_and_Inverse_Thresholds_Source.zip` (24 files, 714,622 bytes unpacked; `Report154.tex`, 1311 lines, 19 pages, dated 3 October 2026); the base. Contributes the exact-saddle expansion of A192563 and of A344639 on compact aspect ratios, the smooth inverse with its equality-safe brackets, and the nested-logarithm inverse.

- **Part II**, bundle Report 155, archive `Matched_Cycles_Exact_Certificates_All_Orders_and_Inverses_Source.zip` (22 files, 923,684 bytes unpacked; `Report155.tex`, 1243 lines, 20 pages, dated 3 October 2026). Contributes the positive Fock certificate and its rational version, the negative-root generalization, the all-orders expansion in  $1/W_0(en)$  with its coefficient generator, and the nested-Lambert inverse.

Where the merge had to choose:

1. *One report rather than two.* The two sequences are two members of one OEIS family and fit one frame (“One family, two methods”); the intake’s alternative, two single-source reports, was not taken. The cost is a heavier notation: the table above separates the letters, and the two Parts are printed whole and independently.
2. *Base and order.* Report 154 first: its object is the two-parameter array, and it has the lower bundle number. There is no dependency either way.
3. *Front matter and bibliography.* The two title blocks, abstracts and scope paragraphs are printed at the heads of their Parts; one table of contents replaces two. The two bibliographies are merged into one, keeping every detail of every entry. Both cite the DLMF, with different details; the entries are kept apart, as [10] (Part I) and [14] (Part II). The write added the entries [18] and [19].
4. *Macros.* Part II’s `\stir` is defined as Part I’s `\sone`; the printed symbol is the same. The other macros the two sources share have the same definitions.
5. *Write additions.* This front matter, the reading-conventions tables of the Parts, the dated notes, Section 23 with Proposition 23.1 and Remark 23.2, labels for the subsections cited, and the label prefixes. Everything else is delivered text.

## What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 107, 5–6 October 2026) both manuscripts were read in full and no step was found false. Checked there by hand: Lemma 3.1 ( $h_N$  nonincreasing, the derivative  $b/r$  of  $rL'_N(r)$ ); Lemma 4.2; the recurrence step of (8.1); the cancellation in (9.4); Part II’s golden bound (14.1), the admissibility of  $q = K/(K + 1)$  for  $K \geq 2$ , the cancellation after (18.9), (19.10) and the directions of the brackets. Checked numerically there:  $\kappa_1 = n$  at the solved saddle and the  $R = 2$  entries of Table 1 at  $n = 20, 100$ ;  $b_{10}$  by the Stirling recurrence; the diagnostics recorded at Proposition 17.1. The write re-read both manuscripts; recomputed the listed terms of A192561, A192562, A192563 and A344639, A192563’s b-file to  $n = 270$ , the identity (2.1) for  $n, k < 30$  and Proposition 23.1 for  $1 \leq n \leq 55$ ; checked Part II’s  $c_2, \dots, c_6$  and (16.3) symbolically, and  $E_1, E_2, E_3, A_2, A_3$  and  $H_1, H_2$  by hand; recomputed Part II’s diagnostic at  $n = 200$  (and 400); and derived the classifications of the inverses above, including the template coefficients  $q_0, q_1$ . The delivered suites were rerun on copies (README).

What was *not* done: the analytic estimates (the contour and weighted Taylor bounds and the parameter derivatives of Part I; the global product bound, the scaled tails, the Gamma-ratio estimate and the inverse windows of Part II) were read and their displayed algebra checked, but they are the manuscripts’ proofs, not an independent verification. The write did not read any of the cited papers or the cited DLMF equations: what is said about Bényi–Ramírez, Komatsu, Jha,

Yakymiv, Komatsu–Szalay, Wilf, Alpay–Porat, Gittenberger–Mandlbürger, Bender–Richmond and Gao–Laferrière–Panario, and the inspection levels recorded, are the manuscripts’. In particular Part I’s novelty statements remain provisional on Yakymiv’s paper [9], as Part I says.

## Relation to neighbouring reports and formal projects

No other report of the repository treats A192561, A192562, A192563, A344639 or poly-Cauchy numbers (repository search, 6 October 2026). Neighbours by object or method only: `a064856-stirling-catalan-transforms` (same batch) studies Stirling transforms of the Catalan numbers; `a122399-surjection-diagonal` gives an all-orders saddle expansion of a two-parameter array uniformly for compact aspect ratios, as Theorem 1.2 does here; `a139383-iterated-bell-diagonals` extracts coefficients  $n![z^n]$  of functions that change with  $n$ , as (2.4) does; `a277364-bell-asymptotics` treats partial row sums of second-kind Stirling numbers; `a124380-signed-moment-asymptotics` treats  $\sum_k z^k \prod_{j \leq k} (1 + jz)$ , a different product of linear factors. None shares a definition, lemma or constant with this report. The transseries volume [19] is related to the inverses as stated above.

Placement in the collection confers no formal status, and no statement of this report is formalized. The two classical identities that Part I’s re-proof of (2.1) combines are formalized in the FabiusFunction Lean layer, as `Fabius.X_pow_eq_sum_stirlingSecond_succ_mul_descPochhammer_comp` (`Analysis/FabiusFunction/Lean/FabiusFunction/StirlingShiftedEvaluations.lean`) and `Fabius.stirlingFirst_succ_succ_eq_sum_choose` (`.../StirlingBasisChange.lean`); (2.1) itself is not.

## Part I

# Diagonal poly-Cauchy permutations at all fixed orders

Bundle Report 154, archive `Diagonal_Poly_Cauchy_All_Orders_and_Inverse_Thresholds_Source.zip`, delivered as `Report154.tex` (19 pages), the base of the merge. Delivered title block “Diagonal poly Cauchy permutations at all fixed orders / Exact saddle expansion and inverse thresholds for OEIS A192563 / Report 154 / 3 October 2026”. Printed as delivered, with `[write]` notes; section, statement, equation and table numbers are those of the delivery.

| Reading conventions of Part I                                                               | Elsewhere in the report                                                                         |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| $a_n$ the sequence A192563; $A_{n,k}$ the array A344639                                     | Part II: $a = a_n$ a saddle point; $A_r$ coefficients                                           |
| $b = \kappa_2$ ; $B_j(p)$ Bernoulli cumulant polynomials                                    | Part II: $b_n = A192561$ , $b$ a centre; $B_n = b_n/(n!)^2$ ; Bernoulli $B_{2j}$ , $B_{m+1}(t)$ |
| $r$ the saddle, $t = e^r$ ; $E_q$ contraction polynomials                                   | Part II: $r$ an index, $t$ a complex variable; $E_r$ Stirling-series coefficients               |
| $L_N(z)$ ; $L = \log X$ ; $F_n$ , $P_R$ , $Y_R$ , $\nu_R$ ; thresholds $T_{\geq}$ , $T_{>}$ | Part II: $L(x)$ , $L = \log T$ ; $F_R$ a logarithmic model; $P_n(z)$ ; $N(T)$                   |

**Abstract of this Part (as delivered).** The diagonal second-kind poly-Cauchy numbers satisfy  $a_n = \sum_m \begin{bmatrix} n \\ m \end{bmatrix} (m+1)^n$  and count the  $(n, n)$ -poly-Cauchy permutations of Bényi and Ramírez. Equivalently,  $a_n/n!$  is the  $n$ th ordinary moment of one plus the cycle count of a uniform permutation of  $n$  labels. We prove a complete asymptotic expansion at every fixed order, using the exact saddle of  $e^z \Gamma(n + e^z)/\Gamma(e^z)$  and explicit finite Gaussian contraction polynomials in its standardized cumulants. The small scale is  $r/n$ , where  $r = \log n - \log \log n - \log \log \log n + o(1)$ . A positive exponential sum controls the whole complementary contour, and a weighted Taylor argument proves the integrated remainder. The result is uniform when the moment order  $k$  and permutation size  $n$  have ratio in a fixed compact subset of  $(0, \infty)$ . Differentiating an explicit real-parameter model then gives arbitrarily fine smooth inverse windows, with strictly justified floor and ceiling bounds at equality cases. We also obtain an explicit nested-logarithm inversion, including its relative remainder. Exact companion calculations check the coefficient identities and the OEIS prefix; optional high-precision diagnostics are separated from the proof.

**Scope and attribution.** The sequence, signed conventions, exact identities, and combinatorial interpretation are established prior art. This report supplies a sequence-specific fixed-order saddle analysis and its inverse consequences. It does not claim exhaustive priority. Several related full texts, notably Yakymiv’s cycle-moment paper, remain uninspected. No numerical asymptotic onset or universal error constant is certified here. No claim concerns growing truncation order or exponentially small sectors.

## 1 Results and mathematical purpose

Let  $\begin{bmatrix} n \\ m \end{bmatrix}$  and  $\begin{Bmatrix} n \\ m \end{Bmatrix}$  be, respectively, the unsigned Stirling numbers of the first kind and the Stirling numbers of the second kind, with both  $(0, 0)$  entries equal to one. Define

$$A_{n,k} = \sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix} (m+1)^k, \quad a_n = A_{n,n} \quad (n, k \geq 0). \quad (1.1)$$

The diagonal begins

$$1, 2, 13, 161, 3148, 87784, 3274640, 156359874.$$

It is OEIS A192563, the diagonal of A344639 [1, 2]. Throughout, logarithms are natural and all asymptotic statements refer to positive arguments tending to infinity.

[write] **6 October 2026.** The live entries, read by the write: A192563 revision #13 (26 May 2021, the date of the mirror snapshot of [1]) and A344639 revision #35 (15 April 2025); they are quoted in the front matter (“The OEIS family”). The eight values above agree with the entry; its b-file ( $0 \leq n \leq 270$ , Stefano Spezia), which the manuscript could not retrieve (Subsection 2.3), agrees with (1.1) at every index (exact recomputation by the write). Part II treats A192561, entered by the same author on the same day in the same pattern; there  $a$  is a saddle point and  $b_n$  the sequence.

## 1.1 Why this diagonal is a natural asymptotic problem

If  $C_n$  is the number of cycles of a uniform random permutation of  $n$  labels, then

$$\frac{A_{n,k}}{n!} = \mathbb{E}(C_n + 1)^k. \quad (1.2)$$

Thus the diagonal studies a moment order that grows with the underlying combinatorial size. The familiar fixed-order cycle-moment regime does not settle it. Large moments give disproportionate weight to atypically large cycle counts, so replacing  $C_n$  by its typical order  $\log n$  is not a justified approximation.

There is also a direct combinatorial class. Take a permutation of  $n$  red labels, put its cycles into canonical order with the largest entry of each cycle first and these largest entries increasing, and assign each of  $k$  blue labels to one of the  $m + 1$  gaps around its  $m$  cycles. Within each gap put the blue labels in increasing order. Removing cycle parentheses produces the second-kind poly-Cauchy permutations of Bényi–Ramírez [3, Theorem 2.1]. This description explains both positivity and the factor  $(m + 1)^k$ . It counts structured permutations on two specified label sets, not arbitrary permutations of their total size.

The present results consequently serve three concrete purposes: asymptotic enumeration of that class, evaluation of growing ordinary cycle-count moments, and inversion of a prescribed counting or moment scale. The compact aspect-ratio theorem permits the two label sets to grow at comparable but unequal rates.

## 1.2 The exact saddle and its coefficients

For real  $N \geq 1$  and real  $z$ , set

$$L_N(z) = z + \log \Gamma(N + e^z) - \log \Gamma(e^z). \quad (1.3)$$

Let  $r = r(N) > 0$  solve

$$rL'_N(r) = N, \quad \text{or} \quad r[1 + e^r\{\psi(N + e^r) - \psi(e^r)\}] = N. \quad (1.4)$$

Here  $\psi$  is the digamma function. Existence and uniqueness for every  $N \geq 1$  are proved in Section 3. Define

$$\kappa_j(N) = (z\partial_z)^j L_N(z) \Big|_{z=r(N)}, \quad b = \kappa_2, \quad h = \frac{r}{N}. \quad (1.5)$$

The differential operator acts with  $N$  held fixed, before evaluation at the saddle. In particular  $\kappa_1 = N$  and  $b > 0$ .

Put  $E_0 = 1$ . For  $q \geq 1$ , define the finite polynomial expression

$$E_q = \sum_{\substack{m_3, \dots, m_{2q+2} \geq 0 \\ \sum_{j=3}^{2q+2} (j-2)m_j = 2q}} (-1)^{J/2} (J-1)!! \prod_{j=3}^{2q+2} \frac{1}{m_j!} \left( \frac{\kappa_j}{j! b^{j/2}} \right)^{m_j}, \quad J = \sum_{j=3}^{2q+2} j m_j. \quad (1.6)$$

The constraint makes  $J$  even; the apparent half-powers of  $b$  thus combine to integer powers in every monomial. We use  $(-1)!! = 1$  when representing  $E_0$  by the empty product.

**Theorem 1.1** (All fixed orders on the diagonal). *For every fixed integer  $R \geq 1$ , as  $n \rightarrow \infty$ ,*

$$a_n = \frac{n! \exp L_n(r)}{r^n \sqrt{2\pi b}} \left( \sum_{q=0}^{R-1} E_q(n) + O_R((r/n)^R) \right). \quad (1.7)$$

Moreover, for each fixed  $j \geq 1$ ,

$$r(N) = \log N - \log \log N - \log \log \log N + o(1), \quad (1.8)$$

$$\kappa_j(N) \sim N r^{j-1}, \quad E_q(N) = O_q((r/N)^q). \quad (1.9)$$

In particular,

$$E_1 = \frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} \sim -\frac{r}{12N}. \quad (1.10)$$

The implicit constants and sufficiently large onsets may depend on  $R$  or on the displayed fixed index.

The leading equivalent already involves an exact transcendental saddle. Retaining it is essential at higher orders. The replacement  $E_1 \sim -r/(12n)$  explains a sign and a scale, but does *not* replace the full  $E_1$  in the  $R = 2$  formula. The errors in that replacement, and in a finite nested-log approximation to  $r$ , can be much larger than  $(r/n)^2$ . This is an exact-saddle expansion with slowly varying coefficients, not a constant-coefficient power series in  $1/n$ .

**Theorem 1.2** (Compact aspect ratios). *Fix  $0 < \alpha \leq \beta < \infty$  and a fixed integer  $R \geq 1$ . For positive integers  $n, k$  with  $\alpha \leq k/n \leq \beta$ , let  $r > 0$  solve  $r L'_n(r) = k$  and define  $\kappa_j, b, E_q$  as in (1.5)–(1.6), with  $n$  held fixed. Uniformly in this range,*

$$A_{n,k} = \frac{k! \exp L_n(r)}{r^k \sqrt{2\pi b}} \left( \sum_{q=0}^{R-1} E_q + O_{R,\alpha,\beta}((r/k)^R) \right). \quad (1.11)$$

Also, uniformly in the same range,

$$r = \log n - \log \log n - \log \log \log n + \log(k/n) + o(1), \quad (1.12)$$

$$\kappa_j \sim k r^{j-1}, \quad b \sim k r. \quad (1.13)$$

The theorem makes no uniform assertion as  $k/n$  approaches zero or infinity.

[write] **6 October 2026.**  $A_{n,k}$  is the entry  $A(n, k)$  of OEIS A344639 (rows  $n$ , columns  $k$ ), so Theorem 1.2 expands that array to every fixed order, uniformly in every sector  $\alpha \leq k/n \leq \beta$ ; Theorem 1.1 is its main diagonal. It says nothing about fixed rows or fixed columns of the array, or about its antidiagonal sums (A344640). See the front matter, “The OEIS family”.

### 1.3 Inverse results and rounding

For fixed  $R \geq 1$ , put

$$P_R(N) = \sum_{q=0}^{R-1} E_q(N), \quad Y_R(N) = \frac{\Gamma(N+1) \exp L_N(r(N))}{r(N)^N \sqrt{2\pi b(N)}} P_R(N). \quad (1.14)$$

This defines an explicit smooth function for sufficiently large real  $N$ . It does not assert an extension of the combinatorial sequence to noninteger indices.

**Theorem 1.3** (Smooth inverse and integer thresholds). *The function  $P_R$  is eventually positive and  $Y_R$  is eventually strictly increasing, with*

$$(\log Y_R)'(N) = 2 \log N - \log r(N) + o(1) \sim 2 \log N. \quad (1.15)$$

For sufficiently large  $X$ , let  $\nu_R$  be the unique solution of  $Y_R(\nu_R) = X$  in the eventual increasing range. There is a constant  $C_R > 0$  such that, with

$$\varepsilon_R = C_R \frac{(\log \nu_R)^{R-1}}{\nu_R^R}, \quad (1.16)$$

the two thresholds

$$T_{\geq}(X) = \min\{n \geq 0 : a_n \geq X\}, \quad T_{>}(X) = \min\{n \geq 0 : a_n > X\}$$

satisfy

$$\lfloor \nu_R - \varepsilon_R \rfloor + 1 \leq T_{\geq}(X) \leq T_{>}(X) \leq \lceil \nu_R + \varepsilon_R \rceil. \quad (1.17)$$

In addition, if  $L = \log X$ ,  $\ell = \log L$ , and  $m = \log \ell$ , then

$$\nu_R = \frac{L}{2\ell} \left[ 1 + \frac{\frac{3}{2}m + 1 + \log 2}{\ell} + O_R \left( \frac{m^2}{\ell^2} \right) \right]. \quad (1.18)$$

The shrinking  $\varepsilon_R$  measures uncertainty around a *smooth* center before rounding. An integer threshold cannot in general equal  $\nu_R + o(1)$  without a rounding qualification. If the interval  $[\nu_R - \varepsilon_R, \nu_R + \varepsilon_R]$  lies strictly between consecutive integers  $j$  and  $j+1$ , both thresholds equal  $j+1$ . If  $X = a_n$ , then  $T_{\geq}(X) = n$  and  $T_{>}(X) = n+1$ ; the two-sided statement (1.17) deliberately retains that possibility. The explicit nested-log expression has a much coarser error than (1.16); it must not silently replace  $\nu_R$  in that fine bracket.

[write] **6 October 2026.** Relation to the transseries volume [19] (front matter, “The inverses and the transseries volume”): the bracket (1.17) rounds a smooth model, not an interpolation of  $(a_n)$ , so it is not an instance of `p0:thm:staircase`; it plays the role of that theorem’s separation condition (2). The nested inverse (1.18) is not an instance of `p0:prop:factorial-core`, because of the term  $-N \log \log N$  in (9.1); its first correction agrees with the first coefficients that `plt:thm:lw-template` gives for a truncated datum, and its remainder is proved here (Section 9).

## 2 Established identities and source boundaries

### 2.1 Signs and the OEIS identification

In Komatsu’s notation for second-kind poly-Cauchy numbers,

$$A_{n,k} = (-1)^n \hat{c}_n^{(-k)}.$$

Bényi–Ramírez write the nonnegative normalization as  $\widehat{c}_{n,k}$ . These conventions exclude the signed first-kind poly-Cauchy numbers, which are a different family [4, Theorem 4]. The final published version of Bényi–Ramírez gives (1.1) and its interpretation in Theorem 2.1 and gives

$$A_{n,k} = \sum_{j=0}^k j! \begin{bmatrix} n+1 \\ j+1 \end{bmatrix} \left\{ \begin{matrix} k+1 \\ j+1 \end{matrix} \right\} \quad (2.1)$$

in Theorem 2.6 [3]. The diagonal of (2.1) is precisely the defining formula in A192563. The theorem numbers in the earlier preprint are different; all theorem references here use the final 2022 volume version.

For completeness, a direct algebraic check of (2.1) is short. The standard identity

$$(m+1)^k = \sum_{j=0}^k \left\{ \begin{matrix} k+1 \\ j+1 \end{matrix} \right\} (m)_j$$

uses the falling factorial  $(m)_j = m(m-1)\cdots(m-j+1)$ . The coefficient of  $x^j$  in

$$\sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix} (1+x)^m = (1+x)(2+x)\cdots(n+x)$$

is  $\begin{bmatrix} n+1 \\ j+1 \end{bmatrix}$ , and also equals  $\sum_m \begin{bmatrix} n \\ m \end{bmatrix} \binom{m}{j}$ . Multiplying by  $j!$  and interchanging finite sums proves the formula. The companion verifies both sides by independent recurrences.

**[write] 6 October 2026.** On the diagonal  $k = n$ , (2.1) is word for word the defining formula of A192563, `Sum_{k=0..n} abs(Stirling1(n+1,k+1))*Stirling2(n+1,k+1)*k!`, so (1.1) agrees with Munarini’s form for every  $n \geq 0$ . The check above is a complete proof. Its standard identity has a one-line proof: dividing  $(m+1)^{k+1} = \sum_{i \geq 1} \left\{ \begin{matrix} k+1 \\ i \end{matrix} \right\} (m+1)_i$  by  $m+1$ , with  $(m+1)_i = (m+1)(m)_{i-1}$ , gives  $(m+1)^k = \sum_{j \geq 0} \left\{ \begin{matrix} k+1 \\ j+1 \end{matrix} \right\} (m)_j$  (the term  $i = 0$  vanishes because  $\left\{ \begin{matrix} k+1 \\ 0 \end{matrix} \right\} = 0$ ). The intake checked the equality for  $n \leq 11$ , the companion checks (2.1) for  $0 \leq n, k \leq 64$ , and the write for  $0 \leq n, k < 30$ . A344639 cites the arXiv preprint of [3] for these formulas, by the preprint’s own theorem numbers. Both classical inputs of the check are formalized in the repository’s FabiusFunction Lean layer (front matter, last section); (2.1) itself is not.

## 2.2 The coefficient function

The rising-factorial identity gives the entire function

$$F_n(z) = \sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix} e^{(m+1)z} \quad (2.2)$$

$$= e^z \prod_{q=0}^{n-1} (e^z + q) = e^z \frac{\Gamma(n + e^z)}{\Gamma(e^z)}. \quad (2.3)$$

The gamma quotient in (2.3) is understood through its finite product wherever necessary;  $F_n$  is entire. Hence

$$A_{n,k} = k! [z^k] F_n(z), \quad a_n = n! [z^n] F_n(z). \quad (2.4)$$

Summing the rising-factorial exponential generating function in  $n$  also gives

$$\sum_{n,k \geq 0} A_{n,k} \frac{x^n y^k}{n! k!} = e^y (1-x)^{-e^y}. \quad (2.5)$$

These are exact prior-art identities, including the shifted-family specialization behind (2.5). The one-variable form (2.4) is the convenient route for the asymptotic analysis; it changes with  $n$ , so the proof must control a triangular family of entire functions.

## 2.3 What the literature screen establishes

The official OEIS data-mirror entry for A192563 was inspected in full on 3 October 2026, as was A344639; the latter was also read at its OEIS page. Neither inspected entry contains a diagonal asymptotic. The individual A192563 webpage and its b-file were not successfully retrieved in that pass. Thus the verified finite prefix used here comes from the official mirror, not from a claimed b-file inspection.

Full inspected primary texts include Komatsu’s 2013 article [4], the final Bényi–Ramírez article [3], Komatsu’s 2021 transform preprint [5], and Jha’s cycle-moment preprint [6]. The first three provide exact identities and related generalizations; no matching diagonal all-orders theorem was located in those texts. Jha’s fixed-order factorial-moment result is not a theorem uniform in the growing ordinary moment order used here.

The following limits matter to any novelty claim. The full texts of Komatsu–Szalay’s shifted-family paper [7], Komatsu’s 2019 recurrence paper [8], and the final 2022 version of the transform paper were not inspected. Most importantly, Yakymiv’s paper on moments of cycle counts of restricted permutations [9] was identified through its official abstract, but its full text and its moment-order uniformity were not verified. This report therefore does not rule out overlap with that paper. A bounded search for the sequence identifiers, diagonal poly-Cauchy asymptotics, and growing cycle-count moments found no matching result; absence from that search is not an exhaustive priority proof.

The analytic inputs below are classical gamma-function formulas [10]. The mathematical contribution presented here is their application with a complete fixed-order error argument, exact coefficient retention, compact aspect-ratio uniformity, and rounding-aware inverse consequences. Source snapshots and precise inspection levels are recorded in the accompanying provenance file.

[write] **6 October 2026.** The write read the live A192563 entry and its b-file (note in Section 1); neither, nor the current A344639, states an asymptotic formula. The provenance file is shipped as `data/154-polycauchy-SOURCE-PROVENANCE.json`. The write did not read any of the cited papers, so the inspection levels stated here, and the caveat about Yakymiv’s paper [9], are the manuscript’s. Part II cites none of these sources; the two manuscripts share only the DLMF.

## 3 Real parameter saddles and scale estimates

This section works with real  $N \geq 1$ . It simultaneously supplies integer saddle estimates and the smooth continuation required for inversion. Write  $t = e^r$  and

$$q_N(t) = t\{\psi(N+t) - \psi(t)\}. \quad (3.1)$$

**Lemma 3.1** (Existence and uniqueness). *For every real  $N \geq 1$ , the equation  $rL'_N(r) = N$  has exactly one solution  $r > 0$ . At this solution  $b > 0$ , and  $r(N)$  is smooth.*

*Proof.* The digamma difference integral from DLMF 5.9.16 [10] gives

$$q_N(t) = \int_0^\infty t e^{-tu} h_N(u) du, \quad h_N(u) = \frac{1 - e^{-Nu}}{1 - e^{-u}}. \quad (3.2)$$

For  $u > 0$ ,

$$\frac{h'_N(u)}{h_N(u)} = \frac{N}{e^{Nu} - 1} - \frac{1}{e^u - 1} \leq 0,$$

because  $e^{Nu} - 1 \geq N(e^u - 1)$  for  $N \geq 1$ . The last inequality follows, for example, from convexity in the exponent parameter. If  $V$  is exponential with rate one, (3.2) is  $\mathbb{E}h_N(V/t)$ . Since  $h_N$  is nonincreasing, this expectation is nondecreasing in  $t$ . The integral is smooth for  $t > 0$ , so  $q'_N(t) \geq 0$ . Consequently

$$L'_N(r) = 1 + q_N(e^r) > 0, \quad L''_N(r) \geq 0.$$

Thus  $rL'_N(r)$  is strictly increasing for  $r > 0$ , begins at zero, and tends to infinity, since it is at least  $r$ . Its derivative is  $L'_N(r) + rL''_N(r) = b/r > 0$ . Existence, uniqueness, and smoothness follow from continuity and the implicit function theorem.  $\square$

**Lemma 3.2** (Derivative bounds with an integral remainder). *For every fixed integer  $j \geq 0$  and  $x > 0$ ,*

$$\left| \frac{d^j}{dx^j} \{ \psi(x) - \log x \} \right| \leq \frac{j!}{x^{j+1}}. \quad (3.3)$$

*In particular, for each fixed  $j \geq 1$ ,*

$$\psi(x) = \log x + O(x^{-1}), \quad \psi^{(j)}(x) = (-1)^{j-1}(j-1)!x^{-j} + O_j(x^{-j-1}). \quad (3.4)$$

*Proof.* The digamma remainder integral, DLMF 5.9.13 [10], is

$$\psi(x) - \log x = - \int_0^\infty G(u) e^{-xu} du, \quad G(u) = \frac{1}{1 - e^{-u}} - \frac{1}{u}. \quad (3.5)$$

For  $u > 0$  one has  $0 < G(u) < 1$ : the lower inequality uses  $1 - e^{-u} < u$ , and the upper one uses  $e^u - 1 > u$ . On each compact subinterval of  $x > 0$ , dominated differentiation is valid to every fixed order. The absolute value of the  $j$ th derivative of (3.5) is at most  $\int_0^\infty u^j e^{-xu} du = j!x^{-j-1}$ . This proves the claims without differentiating an unspecified asymptotic remainder.  $\square$

**Lemma 3.3** (Saddle and cumulant scales). *As  $N \rightarrow \infty$ , with  $t = e^{r(N)}$ ,*

$$\frac{t}{N} \sim \frac{1}{r \log r}, \quad r \sim \log N, \quad r = \log N - \log \log N - \log \log \log N + o(1). \quad (3.6)$$

*For every fixed  $j \geq 1$ ,*

$$L_N^{(j)}(r) = \frac{N}{r} + O_j(t+1) \sim \frac{N}{r}, \quad \kappa_j \sim Nr^{j-1}. \quad (3.7)$$

*Proof.* Lemma 3.2 first yields, uniformly for  $N, t \geq 1$ ,

$$q_N(t) = t \log(1 + N/t) + O(1). \quad (3.8)$$

At  $r = \frac{1}{2} \log N$ , the left side of the saddle equation is  $O(\sqrt{N} \log^2 N) = o(N)$ . At  $r = \log N$  it exceeds  $N$  for large  $N$ , since  $q_N(N) = N \log 2 + O(1)$ . Monotonicity therefore gives

$$\frac{1}{2} \log N < r < \log N \quad (3.9)$$

for all sufficiently large  $N$ .

Put  $x = N/t$ . The saddle equation and (3.8) give

$$\frac{x}{r} = \log(1 + x) + O(t^{-1}). \quad (3.10)$$

Here  $x \geq 1$ ,  $r \rightarrow \infty$ , and  $t \rightarrow \infty$ . A bounded subsequence of  $x$  would contradict (3.10), so  $x \rightarrow \infty$ . Hence  $x/(r \log x) \rightarrow 1$ . The function  $x/\log x$  is eventually increasing. Substitution of fixed multiples of  $r \log r$  into this function brackets  $x$  between such multiples, proving  $\log x \sim \log r$  and then  $x \sim r \log r$ . Taking logarithms once more gives

$$\log x = \log r + \log \log r + o(1).$$

Since  $r = \log N - \log x$ , the refined formula in (3.6) follows from (3.9) and  $r/\log N \rightarrow 1$ .

For derivatives, apply  $(t\partial_t)^{j-1}$  to  $q_N(t)$ . The explicit bounds (3.3) imply that the remainder in (3.8) and each of its fixed  $t\partial_t$  derivatives are  $O_j(1)$ . Indeed each resulting term has a power of  $t$  multiplying a derivative bounded at  $t$  or at  $N+t$  by the corresponding inverse power. In the main term  $t \log(1 + N/t)$ , the original logarithmic term survives once after every application of  $t\partial_t$ ; all other terms are  $O_j(t)$ . It follows that

$$L_N^{(j)}(r) = t \log(1 + N/t) + O_j(t+1) = N/r + O_j(t+1).$$

For  $j = 1$  the saddle equation gives the exact equality  $L'_N(r) = N/r$ . Since  $rt/N \sim 1/\log r \rightarrow 0$ , these expressions are asymptotic to  $N/r$ .

Finally, the Stirling differential-operator identity is

$$(z\partial_z)^j L_N(z) = \sum_{v=1}^j \left\{ \begin{matrix} j \\ v \end{matrix} \right\} z^v L_N^{(v)}(z). \quad (3.11)$$

At  $z = r$ , its term  $v = j$  is asymptotic to  $Nr^{j-1}$  and its other fixed terms are smaller. This proves the cumulant estimate.  $\square$

### 3.1 A finite Bernoulli representation at integer size

For integer  $n$ , the product form gives a second, elementary calculus for the derivatives. Write

$$p_q = \frac{t}{t+q} \quad (0 \leq q \leq n-1). \quad (3.12)$$

Let  $B_j(p)$  be the  $j$ th cumulant polynomial of a Bernoulli variable with success probability  $p$ . These polynomials satisfy

$$B_1(p) = p, \quad B_{j+1}(p) = p(1-p)B'_j(p), \quad B_j(p) = p + O_j(p^2). \quad (3.13)$$

Then

$$L'_n = 1 + \sum_q p_q, \quad L_n^{(j)} = \sum_q B_j(p_q) \quad (j \geq 2). \quad (3.14)$$

Integral comparison shows

$$\sum_q p_q = t \log(1 + n/t) + O(1), \quad \sum_q p_q^2 = O(t+1).$$

This recovers the fixed-derivative estimates without numerical differentiation. It also furnishes the optional diagnostic program with a numerically stable finite-sum route, independent of repeated differentiation of the gamma quotient.

## 4 Contour control and the integrated expansion

Only integer  $n$  is needed in the coefficient integral. All real parameter arguments above serve estimates and the later inverse; we do not integrate a noninteger-size generating function. Fix a small constant  $0 < \rho < \pi/3$ .

**Lemma 4.1** (A fixed complex neighborhood). *For each fixed  $j \geq 1$ , the analytic logarithm of  $F_n$  agreeing with  $L_n$  on the real line satisfies*

$$|L_n^{(j)}(z)| \leq C_j n/r \quad (|z - r| \leq \rho) \quad (4.1)$$

for all sufficiently large  $n$ . If  $f(\theta) = L_n(re^{i\theta})$ , then, on intervals with  $r|\theta| = o(1)$ ,

$$|f^{(j)}(\theta)| \leq C_j n r^{j-1}. \quad (4.2)$$

*Proof.* In the disk  $|z - r| \leq \rho$ , one has  $\operatorname{Re} e^z \geq ce^r = ct$ . Hence every factor  $e^z + q$  is nonzero there, and

$$\left| \frac{e^z}{e^z + q} \right| \leq C \frac{t}{t + q} = Cp_q.$$

The logarithmic derivatives of  $e^z + q$  are the same fixed Bernoulli cumulant polynomials as in (3.13), evaluated at  $e^z/(e^z + q)$ . Each has zero constant coefficient. The sum of their absolute values is bounded by a fixed multiple of  $\sum_q p_q = O(n/r)$ ; the extra term  $z$  is harmless. A nonvanishing analytic function on this disk has an analytic logarithm, establishing (4.1). Applying the chain rule repeatedly, or (3.11), gives (4.2).  $\square$

Set  $\delta = n^{-2/5}$ . At  $\theta = 0$ ,  $f^{(j)}(0) = i^j \kappa_j$ . Taylor's theorem through degree two therefore gives, uniformly for  $|\theta| \leq \delta$ ,

$$\begin{aligned} \operatorname{Re}\{L_n(re^{i\theta}) - L_n(r) - in\theta\} &\leq -\frac{1}{2}b\theta^2 + Cnr^2|\theta|^3 \\ &\leq -\frac{1}{4}b\theta^2. \end{aligned} \quad (4.3)$$

The last inequality follows from  $b \sim nr$  and  $r\delta \rightarrow 0$ .

**Lemma 4.2** (The full complementary circle). *For some fixed  $c > 0$  and all large  $n$ ,*

$$\sup_{\delta \leq |\theta| \leq \pi} \frac{|F_n(re^{i\theta})|}{F_n(r)} \leq e^{-cn\delta^2} = e^{-cn^{1/5}}. \quad (4.4)$$

*Proof.* The positive exponential sum (2.2) implies

$$\begin{aligned} |F_n(re^{i\theta})| &\leq \sum_m \begin{bmatrix} n \\ m \end{bmatrix} e^{(m+1)r \cos \theta} \\ &= F_n(r \cos \theta) \leq F_n(r \cos \delta) \quad (\delta \leq |\theta| \leq \pi). \end{aligned}$$

The function  $F_n$  is increasing on the entire real line because all its exponential-sum coefficients are nonnegative and their exponents are positive. On  $s \in [r \cos \delta, r]$ , one has  $s - r = o(1)$ . The finite-sum estimates (3.14) give  $L'_n(s) \sim n/r$  uniformly there. Since  $r - r \cos \delta \asymp r\delta^2$ ,

$$L_n(r \cos \delta) - L_n(r) = - \int_{r \cos \delta}^r L'_n(s) ds \leq -cn\delta^2.$$

This proves the assertion over the whole complementary circle.  $\square$

The last argument is useful because  $e^z$  has an imaginary period, which can make an informal local saddle calculation look vulnerable to secondary peaks. The bound handles the entire circle at once; no alleged periodic peak has been omitted.

#### 4.1 Normalization and the central cutoff

Cauchy's coefficient formula is

$$\frac{a_n r^n}{n! F_n(r)} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp\{L_n(re^{i\theta}) - L_n(r) - in\theta\} d\theta, \quad (4.5)$$

where the integrand on the full circle is interpreted as  $F_n(re^{i\theta})e^{-in\theta}/F_n(r)$ , so a global logarithm is unnecessary. Only the local arc uses its chosen analytic logarithm. After multiplying by  $\sqrt{2\pi b}$ , Lemma 4.2 bounds the complementary contribution by

$$O(\sqrt{b} e^{-cn^{1/5}}),$$

which is smaller than every fixed power of  $r/n$ .

Write

$$\eta = \sqrt{r/n}, \quad u = \sqrt{b}\theta, \quad \lambda_j = \frac{\kappa_j}{b^{j/2}}. \quad (4.6)$$

Then  $\lambda_j = O_j(\eta^{j-2})$  for every fixed  $j \geq 3$ . Restrict further to  $|u| \leq U = \eta^{-1/6}$ . This interval lies inside  $|\theta| \leq \delta$ , since

$$\frac{U}{\sqrt{b}} = O(n^{-5/12} r^{-7/12}) = o(n^{-2/5}).$$

By (4.3), the omitted part of the local arc has normalized integral  $O(e^{-cU^2})$ . This too is smaller than any fixed power of  $\eta$ .

#### 4.2 A weighted remainder that can be integrated

Fix  $R \geq 1$ . Taylor expansion through degree  $2R + 2$ , using Lemma 4.1 at degree  $2R + 3$ , gives

$$L_n(re^{i\theta}) - L_n(r) - in\theta = -u^2/2 + Q(u) + \mathcal{R}(u), \quad (4.7)$$

where

$$Q(u) = \sum_{j=3}^{2R+2} \frac{\lambda_j (iu)^j}{j!}, \quad |\mathcal{R}(u)| \leq C_R \eta^{2R+1} |u|^{2R+3}. \quad (4.8)$$

Indeed the normalized derivative bound is  $n r^{2R+2} / b^{(2R+3)/2} = O_R(\eta^{2R+1})$ . On  $|u| \leq U$ , the bounds imply  $|Q(u)| \leq C_R \eta |u|^3 = o(1)$  and  $|\mathcal{R}(u)| = o(1)$ . Therefore

$$|e^{Q(u)+\mathcal{R}(u)} - e^{Q(u)}| \leq C_R |\mathcal{R}(u)|,$$

and its integral against  $e^{-u^2/2}$  is  $O_R(\eta^{2R+1})$ .

For the remaining exponential, write  $\lambda_j = \eta^{j-2} c_j$ , where all the finitely many  $c_j$  are bounded uniformly in  $n$ . Introduce an auxiliary real variable  $s$  and set

$$H(s, u) = \exp \left( \sum_{j=3}^{2R+2} \frac{c_j (iu)^j s^{j-2}}{j!} \right). \quad (4.9)$$

The exponent is bounded uniformly for  $0 \leq s \leq \eta$  and  $|u| \leq U$ . Differentiating  $H$  exactly  $2R$  times with respect to  $s$  produces  $H$  times a finite polynomial in  $u$ , the bounded  $c_j$ , and nonnegative powers of  $s$ . Thus there is a fixed integer  $M_R$  for which Taylor's theorem in  $s$  yields

$$H(\eta, u) = \sum_{d=0}^{2R-1} \eta^d H_d(u) + O_R(\eta^{2R}(1 + |u|^{M_R})). \quad (4.10)$$

The bound is integrable against the Gaussian, uniformly in  $n$ . This is the necessary integrated error argument. A pointwise Taylor bound evaluated only at the outer cutoff would not by itself prove the asserted relative remainder.

Every monomial of weighted degree  $d$  has ordinary degree  $J = \sum j m_j$  in  $u$ , while  $d = \sum (j-2)m_j$ . Hence  $J \equiv d \pmod{2}$ . Odd  $d$  terms integrate to zero on the symmetric interval. For even  $d = 2q$ , extending each polynomial Gaussian integral from  $[-U, U]$  to  $\mathbb{R}$  incurs only an exponentially decaying error. The identities

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-u^2/2} u^{2v} du = (2v-1)!!, \quad \int_{\mathbb{R}} e^{-u^2/2} u^{2v+1} du = 0 \quad (4.11)$$

give precisely the contraction formula (1.6). Combining (4.5)–(4.10) proves (1.7) with error  $O_R(\eta^{2R}) = O_R((r/n)^R)$ . Each monomial of  $E_q$  is  $O_q(\eta^{2q})$ , proving its size claim. Finally, substituting  $\kappa_j \sim nr^{j-1}$  and  $b \sim nr$  into (1.10) gives  $(1/8 - 5/24)r/n = -r/(12n)$ . This completes the proof of Theorem 1.1.

## 5 Exact coefficient calculus

The coefficients are finite universal expressions in the exact cumulants. For example, the next one after (1.10) is

$$\begin{aligned} E_2 = & -\frac{\kappa_6}{48b^3} + \frac{7\kappa_3\kappa_5}{48b^4} + \frac{35\kappa_4^2}{384b^4} \\ & - \frac{35\kappa_3^2\kappa_4}{64b^5} + \frac{385\kappa_3^4}{1152b^6}. \end{aligned} \quad (5.1)$$

No limiting replacement of a cumulant is part of this formula.

### 5.1 Two independent finite generation rules

Formula (1.6) can be generated by enumerating the nonnegative integer solutions of  $\sum_{j \geq 3} (j-2)m_j = 2q$ . There is also an algebraically independent formal exponential recurrence. For indeterminates  $x_3, x_4, \dots$ , set

$$Q_w(u) = \frac{x_{w+2}(iu)^{w+2}}{(w+2)!}, \quad \exp\left(\sum_{w \geq 1} Q_w(u)s^w\right) = \sum_{d \geq 0} H_d(u)s^d. \quad (5.2)$$

Formal differentiation with respect to  $s$  gives

$$H_0(u) = 1, \quad H_d(u) = \frac{1}{d} \sum_{w=1}^d w Q_w(u) H_{d-w}(u). \quad (5.3)$$

Integrating  $H_{2q}$  term by term against the standard Gaussian and then putting  $x_j = \kappa_j/b^{j/2}$  yields  $E_q$ . Every step takes place in a finite rational polynomial ring after a degree cutoff; convergence of the formal infinite series is not asserted or needed. The companion compares this recurrence with the weighted-partition formula using exact rational arithmetic.

For the Bernoulli polynomials there are likewise two exact rules: (3.13) and

$$B_j(p) = \sum_{v=1}^j (-1)^{v-1} (v-1)! \left\{ \begin{matrix} j \\ v \end{matrix} \right\} p^v. \quad (5.4)$$

The latter follows by expanding  $\log(1 + p(e^z - 1))$  in powers of  $p$  and extracting the  $j$ th derivative at zero. Comparison of these two constructions checks the derivative input used by the numerical diagnostics.

## 5.2 What an exact finite check certifies

The rational contraction identities, Stirling recurrences, Bernoulli identities, and agreement of finite sequence values can all be certified by terminating exact calculations. They check signs, factorial normalizations, parity, and the correct sequence convention. They do not prove the uniform contour estimate or an asymptotic remainder. Those claims depend on the analytic arguments in Sections 3 and 4.

Similarly, a small numerical residual at a finite list of sizes is not an interval bound for all larger sizes. The optional numerical program is explicitly a diagnostic. In particular, it cannot turn the asymptotic threshold theorem into a certified finite- $X$  rounding oracle without explicit constants and an onset.

## 6 Uniformity for comparable moment orders

We give the details needed for Theorem 1.2, rather than merely substituting  $k$  for  $n$  in the diagonal formula. Let  $\lambda = k/n \in [\alpha, \beta]$  and choose the unique positive saddle satisfying  $rL'_n(r) = k$ . The same existence proof applies, since  $rL'_n(r)$  increases from zero to infinity.

The preliminary comparison at  $r = \frac{1}{2} \log n$  and  $r = \log n$  now gives

$$\frac{1}{2} \log n < r < \log n$$

for all large  $n$ , uniformly in  $\lambda$ . Put  $t = e^r$  and  $x = n/t$ . The saddle relation becomes

$$\frac{\lambda x}{r} = \log(1 + x) + O(t^{-1}). \quad (6.1)$$

Since  $\lambda$  stays bounded above and away from zero,  $x \rightarrow \infty$  uniformly and inversion gives

$$x \sim \frac{r}{\lambda} \log r, \quad \log x = \log r + \log \log r - \log \lambda + o(1),$$

uniformly. Taking  $r = \log n - \log x$  proves (1.12). The derivative estimates become

$$L_n^{(j)}(r) = k/r + O_j(t+1) \sim k/r, \quad \kappa_j \sim kr^{j-1}.$$

The constants can be chosen uniformly on the compact ratio interval.

The disk  $|z - r| \leq \rho$  used in Lemma 4.1 is unchanged. The derivative bound is now  $C_j k r^{j-1}$  for the angular function. With  $\delta = n^{-2/5}$ , positivity gives a complementary-circle bound  $\exp(-ck\delta^2)$ , where  $c > 0$  can be chosen uniformly in  $\lambda$ . Since  $k \asymp n$ , this is at most  $\exp(-c'n^{1/5})$ . The central Gaussian argument uses  $\eta = \sqrt{r/k}$  and  $U = \eta^{-1/6}$ ; both the cutoff inclusion and every fixed-degree integrated remainder remain uniform. Applying Cauchy's formula to  $A_{n,k} = k![z^k]F_n(z)$  proves (1.11).

This theorem leaves two different boundary regimes open. Fixed or slowly growing  $k$  is not covered by a compact positive ratio; neither is  $k/n \rightarrow \infty$ . A theorem connecting these regimes would need new uniform estimates, not an unqualified appeal to (1.11).

## 7 Differentiating the smooth model

We now prove the derivative claim needed for inversion. All derivatives in this section are of exact gamma, polygamma, and finite cumulant expressions. No derivative is taken of the asymptotic error in (1.7).

**Lemma 7.1** (Parameter derivatives). *For fixed  $j \geq 1$ , along the real saddle,*

$$r_N \sim N^{-1}, \quad \frac{d\kappa_j}{dN} = O_j(r^{j-1}), \quad \frac{b_N}{b} = O(N^{-1}). \quad (7.1)$$

For fixed  $q \geq 1$ ,

$$E'_q(N) = O_q\left(\frac{(r/N)^q}{N}\right). \quad (7.2)$$

*Proof.* With  $r$  temporarily held fixed, direct differentiation gives

$$\partial_N L_N^{(j)}(r) = (t\partial_t)^j \psi(N+t) = \sum_{v=1}^j \left\{ \begin{matrix} j \\ v \end{matrix} \right\} t^v \psi^{(v)}(N+t) = O_j(t/N). \quad (7.3)$$

The last bound follows from Lemma 3.2, since  $t/N \rightarrow 0$ . Implicit differentiation of (1.4) gives exactly

$$r_N = \frac{r\{1 - rt\psi'(N+t)\}}{b}. \quad (7.4)$$

Now  $rt\psi'(N+t) \sim rt/N \sim 1/\log r \rightarrow 0$  and  $b \sim Nr$ , proving  $r_N \sim 1/N$ .

From (3.11), (3.7), and (7.3), with the other variable fixed,

$$\partial_r \kappa_j = O_j(Nr^{j-1}), \quad \partial_N \kappa_j = O_j(r^j t/N).$$

The chain rule and (7.4) give

$$\frac{d\kappa_j}{dN} = \partial_N \kappa_j + r_N \partial_r \kappa_j = O_j(r^{j-1}),$$

because  $rt/N = O(1)$ . Division by  $\kappa_j \sim Nr^{j-1}$  shows that every fixed  $\kappa_j$  is eventually positive with logarithmic derivative  $O_j(1/N)$ . The same applies to  $b$ .

Differentiate each monomial of (1.6) separately. Its size is  $O_q((r/N)^q)$ , and the product rule multiplies that bound by  $O_q(1/N)$ . Summing finitely many terms proves (7.2). This argument does not assume that the sum  $E_q$  itself is nonzero.  $\square$

It follows that

$$P_R = 1 + O_R(r/N), \quad P'_R = O_R(r/N^2). \quad (7.5)$$

For  $R = 1$ , the derivative is exactly zero. Thus  $P_R > 0$  for large  $N$ . In the logarithm of (1.14), all terms involving  $r_N$  from  $L_N(r) - N \log r$  cancel by the saddle equation. The exact derivative is

$$(\log Y_R)' = \psi(N+1) + \psi(N+t) - \log r - \frac{b_N}{2b} + \frac{P'_R}{P_R}. \quad (7.6)$$

Lemma 3.2 and (7.5) yield

$$(\log Y_R)' = 2 \log N - \log r + O_R(t/N + 1/N) = 2 \log N - \log r + o(1). \quad (7.7)$$

This proves eventual strict monotonicity. It also shows  $Y_R(N) \rightarrow \infty$ , so every sufficiently large  $X$  has one solution in the eventual increasing range. There is no need to assert global monotonicity of  $Y_R$  at small  $N$ .

## 8 Threshold windows and strict equality edges

We first check exact monotonicity of the original sequence. The unsigned Stirling recurrence gives, for  $n \geq 1$ ,

$$a_{n+1} = \sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix} \{n(m+1)^{n+1} + (m+2)^{n+1}\} > a_n. \quad (8.1)$$

Also  $a_1 = 2 > a_0 = 1$ . Since  $a_n \geq n!$ , the sequence tends to infinity, and both thresholds are well defined.

[write] **6 October 2026.** The step behind (8.1) is  $\begin{bmatrix} n+1 \\ m \end{bmatrix} = n \begin{bmatrix} n \\ m \end{bmatrix} + \begin{bmatrix} n \\ m-1 \end{bmatrix}$ . Applied to (2.1) instead, the same recurrence gives an identity for the third member of the family, A192562 (Proposition 23.1).

Theorem 1.1 and  $P_R = 1 + O_R(r/n)$  imply

$$\log a_n - \log Y_R(n) = O_R\left((r(n)/n)^R\right) = O_R\left((\log n/n)^R\right). \quad (8.2)$$

Fix a sufficiently large  $X$ , abbreviate  $\nu = \nu_R$ , and take

$$n_- = \lfloor \nu - \varepsilon \rfloor, \quad n_+ = \lceil \nu + \varepsilon \rceil, \quad \varepsilon = C_R \frac{(\log \nu)^{R-1}}{\nu^R}. \quad (8.3)$$

Both integers are asymptotic to  $\nu$  and, for large  $X$ , lie beyond every onset used in the proof. On the short intervals joining them to  $\nu$ , (7.7) is at least  $\log \nu$ . Consequently

$$\begin{aligned} \log Y_R(\nu) - \log Y_R(n_-) &\geq \varepsilon \log \nu, \\ \log Y_R(n_+) - \log Y_R(\nu) &\geq \varepsilon \log \nu. \end{aligned} \quad (8.4)$$

The local errors in (8.2) are bounded by a constant multiple of  $(\log \nu/\nu)^R$ . Choose  $C_R$  strictly larger than that constant, increasing the onset if necessary. Since  $\varepsilon \log \nu = C_R (\log \nu/\nu)^R$ , this gives the *strict* inequalities

$$a_{n_-} < X < a_{n_+}. \quad (8.5)$$

Exact monotonicity now proves (1.17). A fixed initial prefix causes no difficulty: increasing  $X$  past its largest entry excludes it automatically.

## 8.1 Why the strict inequalities are necessary

The lower edge is  $n_- + 1$ , not merely  $n_-$ . The upper edge bounds the strict threshold because  $X < a_{n_+}$ , not just  $X \leq a_{n_+}$ . If  $X = a_j$ , the threshold definitions give  $j$  and  $j + 1$  respectively. Any derivation which tacitly identifies these two thresholds at equality would be false. The strictly larger error constant in (8.4) ensures that (1.17) covers this case as written.

The theorem supplies a hierarchy of centers. At  $R = 1$ , the smooth uncertainty is  $O(\nu^{-1})$ ; at  $R = 2$  it is  $O(\log \nu / \nu^2)$ ; at  $R = 3$  it is  $O((\log \nu)^2 / \nu^3)$ . The constants are not numerically certified here. These are asymptotic precision statements, not precomputed finite-input guarantees. A certified implementation would require effective constants for the contour and central errors together with interval evaluation of the saddle and inverse.

## 9 First corrected nested logarithm inversion

The fine inverse center is defined implicitly, but its leading nested-logarithm form can be proved with an explicit relative remainder. We begin with the required logarithmic growth estimate.

**Lemma 9.1** (Logarithmic size of the smooth model). *For every fixed  $R \geq 1$ ,*

$$\log Y_R(N) = 2N \log N - N \log \log N - 2N + O_R \left( \frac{N \log \log N}{\log N} \right). \quad (9.1)$$

*Proof.* The logarithmic Stirling formula with its positive-real remainder bound [10, 5.11.1 and 5.11(ii)], together with  $t/N \rightarrow 0$ , gives

$$\log \Gamma(N + t) - \log \Gamma(t) = N \log N - N + t \log(N/t) + O(t + t^2/N + \log N). \quad (9.2)$$

For example, expand  $(N+t) \log(N+t)$  around  $N$  using  $\log(1+t/N) = t/N + O((t/N)^2)$ ; the remaining half-logarithm terms are  $O(\log N)$  and the Stirling remainders are bounded. The saddle relation yields  $t \log(1 + N/t) = N/r + O(1)$  and hence  $t \log(N/t) = O(N/r)$ . Also  $t + t^2/N + \log N = O(N/r)$ . Adding the initial term  $r$  in (1.3) therefore gives

$$L_N(r) = N \log N - N + O(N/r).$$

Now use  $\log \Gamma(N + 1) = N \log N - N + O(\log N)$ ,  $\log b = O(\log N)$ , and  $\log P_R = O_R(r/N)$  in (1.14). We obtain

$$\log Y_R(N) = 2N \log N - N \log r - 2N + O_R(N/r). \quad (9.3)$$

From (1.8),  $\log r = \log \log N + O(\log \log N / \log N)$ . Substitution proves (9.1).  $\square$

Let  $L = \log X$ ,  $\ell = \log L$ , and  $m = \log \ell$ . Equation (9.1) first implies  $\nu_R \sim L/(2\ell)$ : it gives  $L \sim 2\nu_R \log \nu_R$ , so  $\log \nu_R \sim \log L$  and then the stated equivalent. Set

$$c = \frac{3}{2}m + 1 + \log 2, \quad x_0 = \frac{L}{2\ell} \left( 1 + \frac{c}{\ell} \right). \quad (9.4)$$

Elementary logarithmic expansions give

$$\log x_0 = \ell - m - \log 2 + c/\ell + O(m^2/\ell^2), \quad (9.5)$$

$$\log \log x_0 = m - (m + \log 2)/\ell + O(m^2/\ell^2). \quad (9.6)$$

On substituting (9.5)–(9.6) into (9.1), the coefficient of  $L/\ell$  cancels:

$$c - \frac{3}{2}m - 1 - \log 2 = 0.$$

The result is

$$\log Y_R(x_0) = L + O_R(Lm^2/\ell^2). \quad (9.7)$$

Both  $x_0$  and  $\nu_R$  are asymptotic to  $L/(2\ell)$ . Throughout the interval between them, (1.15) is of order  $\ell$ . The mean value theorem applied to (9.7) consequently gives

$$\nu_R - x_0 = O_R(Lm^2/\ell^3).$$

This is exactly (1.18), including its relative  $O_R(m^2/\ell^2)$  remainder, and completes the proof of Theorem 1.3.

The coefficient  $3/2$  has two contributions: inversion of  $N \log N$  contributes one factor of  $\log \log L$ , and the term  $-N \log \log N$  contributes an additional half after normalization by  $2N \log N$ . The constants  $1 + \log 2$  likewise come from the  $-2N$  term and the leading factor two. This explanation is an interpretation of the explicit cancellation, not a substitute for its remainder calculation.

[write] **6 October 2026.** With (8.2), Lemma 9.1 gives  $\log a_n = 2n \log n - n \log \log n - 2n + O(n \log \log n / \log n)$ . Part II's (19.2) is  $\log b_n = 2n \log n - 2n + O((\log n)^2)$ , so  $\log(a_n/b_n) = -n \log \log n + O(n \log \log n / \log n)$ : the two sequences share the factorial core  $2N \log N - 2N$ , and Part II's leading inverse (19.3) is the exact inverse of that core. Because of the extra term  $-N \log \log N$ , Part I inverts by nested logarithms (1.18) rather than through that core.

## 10 Reproducible exact checks and numerical diagnostics

The accompanying archive contains this PDF, its standalone LaTeX source, deterministic build and packaging scripts, source provenance, and a companion directory. It does not require a private proof file, an earlier report, or a live network connection to verify its exact identities. Only the optional diagnostics require an external Python package, pinned as `mpmath==1.3.0`.

The exact companion checks are designed around independent routes:

1. Generate unsigned first-kind and second-kind Stirling rows by their integer recurrences, and compare the moment sum with (2.1) and a derivative-product evaluation of (2.3).
2. Reproduce the entire 17-term prefix in the inspected official A192563 mirror snapshot, with offset zero and the specified sign normalization.
3. Compare weighted-partition Gaussian contractions with the formal exponential recurrence (5.3), using exact rational coefficients. Check the displayed  $E_1$  and  $E_2$  separately.
4. Compare Bernoulli cumulant polynomials generated by (3.13) and (5.4), and verify the Stirling differential-operator identity at finite degrees.
5. Check strict and nonstrict threshold definitions at exact finite equality cases, and exercise the output-safety rules under both ordinary Python and `python3 -O`.

The released exact run checks all 4225 pairs  $0 \leq n, k \leq 64$  by the three integer routes and constructs  $E_0, \dots, E_6$ . Their monomial counts are 1, 2, 5, 11, 22, 42, 77. It performs 11993 individual finite

checks, including the polynomial and threshold identities; normal and optimized runs have the same mathematical payload. No correctness condition relies on Python’s removable `assert` statements. The companion’s documented finite cutoffs are verification parameters, not hypotheses of the theorems. The companion README lists the commands; result files record the checks.

Optional diagnostics solve the real saddle, evaluate cumulants by finite Bernoulli sums, and compare the leading and corrected saddle models with exact integer counts. They report the normalized error

$$\frac{a_n/M_n - \sum_{q=0}^{R-1} E_q}{(r/n)^R}, \quad M_n = \frac{n!F_n(r)}{r^n \sqrt{2\pi b}}, \quad (10.1)$$

and analogous compact-aspect-ratio samples. These values are floating-point observations, separated from exact pass/fail identities and from the proof of the remainder. The diagnostics do not infer rigorous interval bounds or an asymptotic onset from sampled residuals.

Table 1: Observed scaled residuals in (10.1). These rounded 100-digit calculations are diagnostics, not error bounds.

| $n$  | $R = 2$      | $R = 3$      | $R = 4$     |
|------|--------------|--------------|-------------|
| 20   | -0.008764515 | -0.001372810 | 0.000515071 |
| 50   | -0.006294384 | 0.000461294  | 0.001454634 |
| 100  | -0.004675041 | 0.001130859  | 0.001471817 |
| 200  | -0.003449208 | 0.001464946  | 0.001341716 |
| 500  | -0.002300129 | 0.001667662  | 0.001143029 |
| 1000 | -0.001672183 | 0.001739809  | 0.001010160 |

The numerical file additionally covers aspect ratios 1/2 and 2 at  $n = 50, 200, 500$ , and all corrections through  $E_6$ . Changes of sign in the scaled residuals are compatible with the theorem: it supplies a fixed-order magnitude estimate, not a prescribed sign or a monotone approach to a limit.

[write] **6 October 2026.** Shipped names: the companion is `code/154-polycauchy-companion-*.py` with its inputs and results in `data/154-polycauchy-companion-*`, its README is `154-polycauchy-companion-README.md`, and the release scripts are `code/154-polycauchy-*.py`; `Report154.pdf` and `SHA256SUMS` are not shipped. At the write (Python 3.14.4, Windows, fresh extraction from the arrival commit), `exact_checks.py` reproduced `results/exact.json` and `exact-optimized.json`, and `numerical_diagnostics.py` (mpmath 1.3.0) reproduced `numerical.json`, byte for byte after removal of carriage returns; of the 33 tests, 23 passed and the 10 output-safety tests stopped with “POSIX O\_NOFOLLOW is required” (the companion README requires POSIX). An independent computation at intake reproduced the  $R = 2$  entries of Table 1 at  $n = 20$  and  $n = 100$  to every printed digit, and found  $E_1/(-r/12n) = 1.226$  and  $1.154$  there: the slow convergence against which the manuscript warns after Theorem 1.1.

## 10.1 Build and release boundaries

The scripts use exclusive new output paths and reject static symlinked path components. Builds assume a trusted parent directory with no concurrent malicious path replacement; directory creation is not a security boundary against a same-user process swapping the new leaf before it is opened. PDF builds use a fresh directory, fixed metadata, no shell escape, and stable cross-references. The release packager reads a fixed allowlist, checks every member against the SHA-256 manifest, and creates a stored ZIP with fixed timestamps, ordering, and permissions. Byte reproducibility of the

PDF is checked in clean builds under the recorded TeX toolchain; a different TeX or font version may produce different PDF bytes. Exact mathematical checks need only the Python standard library.

No publication, OEIS edit, repository modification, or external submission is part of this report. The source references are public; the archive distributes original exposition and programs, not copies of the cited journal PDFs or private review material.

## 11 Limits and further research

The results cover each fixed truncation order with a proved integrated remainder. They do not provide estimates uniform in  $R \rightarrow \infty$ , an optimal truncation theorem, or a decomposition into exponentially small asymptotic sectors. The fact that the noncentral contour is smaller than every fixed algebraic order does not, by itself, supply any of those stronger claims.

Several extensions have concrete mathematical content:

1. **Effective remainder constants.** Can one turn the disk, contour, and weighted Taylor bounds into practical explicit constants and onsets? Combining those with interval gamma and polygamma evaluation could produce certified finite- $X$  threshold decisions.
2. **Transition between moment regimes.** Determine a uniform expansion as  $k/n \rightarrow 0$ , connecting fixed-order cycle moments to the present growing-order regime. The transition as  $k/n \rightarrow \infty$  is a separate problem; it may involve substantially different saddle scales.
3. **Cycle statistics under the moment weight.** For the probability proportional to  $\lfloor \frac{n}{m} \rfloor (m+1)^k$ , obtain a limit law and local estimates for  $m$ . This would describe the cycle counts typical of large poly-Cauchy structures rather than just their total number. No such limit law is asserted here.
4. **Further explicit inverse terms.** Sharpen (9.3) using higher saddle-scale and gamma expansions, and derive a full hierarchy in nested logarithms with proved remainders. Its error should remain distinct from the much finer exact-saddle inverse window.
5. **Restricted and shifted families.** Investigate restricted cycle lengths and shifted poly-Cauchy parameters after a complete comparison with Yakymiv and the shifted-family literature. Positivity may still control global contours, but the local derivative scales require separate proofs.
6. **Large truncation orders.** Study the growth of the contraction polynomials and their analytic remainder constants. A valid growing-order or exponential-improvement theorem would need information beyond the fixed-order estimates established here.

These questions are proposed research directions, not claims of proved results or of absence from all prior literature. In particular, reading the uninspected cycle-moment source is a priority before making stronger originality statements about restricted permutations or growing moments.

[write] 6 October 2026. Section 23 collects these questions with those of Part II under Vladimir's standing rule. Question 2 concerns the fixed rows and columns of A344639, which Theorem 1.2 does not reach; question 3 has a counterpart in Part II (Subsection 22.4).

## Part II

# Exact positive certificates and all orders for matched cycles

Bundle Report 155, archive `Matched_Cycles_Exact_Certificates_All_Orders_and_Inverses_Source.zip`, delivered as `Report155.tex` (20 pages). Delivered title block “Exact positive certificates and all orders for matched cycles / Factorial weighted squared Stirling numbers in OEIS A192561 / Report 155 / 3 October 2026”. Printed as delivered, with `[write]` notes. Delivered Section  $k$  is Section  $k + 11$  here, its statement or equation  $k.j$  is  $(k + 11).j$ , and its Table 1 is Table 2. Section 23, which closes the report, is added in the write.

| Reading conventions of Part II                                                        | Elsewhere in the report                                        |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------|
| $b_n$ the sequence A192561; $B_n = b_n/(n!)^2$ ; $b$ a centre                         | Part I: $b = \kappa_2$ ; $B_j(p)$ cumulant polynomials         |
| $a = a_n$ the saddle point of (12.2); $A_r$ coefficients                              | Part I: $a_n = \text{A192563}$ ; $A_{n,k} = \text{A344639}$    |
| $w = W_0(en)$ ; $E_r$ , $E(x)$ Stirling-series coefficients; $H_r$ , $\mathcal{H}(w)$ | Part I: $E_q$ contraction polynomials; $H(s, u)$ , $H_d(u)$    |
| $L(x) = -\log(1 - x)$ , $L = \log T$ ; $F_R$ a logarithmic model; $N(T)$              | Part I: $L_N(z)$ , $L = \log X$ ; $Y_R$ ; $T_{\geq}$ , $T_{>}$ |

**Abstract of this Part (as delivered).** For  $b_n = \sum_{k=0}^n \frac{[n+1]_k^2}{k!} k!$ , we give an exact positive expansion with an explicit tail valid at every  $n \geq 1$ . It is obtained by translating the classical Gaussian Fock norm of  $\prod_{j=1}^n (1 + z/j)$  to a positive real center. The saddle moments satisfy the uniform bound  $S_2 \leq 1/\varphi$ , where  $\varphi = (1 + \sqrt{5})/2$ , yielding a geometrically convergent certificate with a universal choice of truncation parameter. A rational-center version makes the certificate computable using rational arithmetic and elementary exponential bounds. The companion isolates the exact integer  $b_{10} = 1273270076895432$  in this way.

For every fixed  $R \geq 1$ , with  $w = W_0(en)$ , we also prove

$$b_n = \frac{(n!)^2 e^{w^2+1}}{2\pi w} \left( \sum_{r=0}^{R-1} \frac{A_r}{w^r} + O_R(w^{-R}) \right),$$

where  $A_0 = 1$ ,  $A_1 = -1/6$ ,  $A_2 = 55/72$ , and  $A_3 = 1381/6480$ . An exact rational recurrence generates every fixed order; values through  $A_{10}$  are supplied. The proof of the all-orders remainder uses a scaled Fock tail and finitely many coefficients. We derive explicit nested Lambert inverses with safe two-ceiling threshold brackets and extend the exact certificate to polynomials whose zeros are strictly negative real numbers.

**Scope and priority.** Fock orthogonality, shifted Gamma expansions, and saddle methods are classical. Wilf’s unweighted squared-Stirling asymptotic is a close antecedent. No sequence-specific formula of the present form was located in the bounded source search; scholarly priority is not established. Constants and starting indices for the inverse-logarithm remainders are not numerically supplied here. The finite- $n$  positive certificate, in contrast, has fully explicit bounds.

## 12 Sequence and principal conclusions

Throughout,  $[m_j]$  denotes an unsigned Stirling number of the first kind. Define

$$b_n = \sum_{k=0}^n \left[ \begin{matrix} n+1 \\ k+1 \end{matrix} \right]^2 k!, \quad P_n(z) = \prod_{j=1}^n \left( 1 + \frac{z}{j} \right), \quad B_n = \frac{b_n}{(n!)^2}. \quad (12.1)$$

An empty product is 1, so  $b_0 = B_0 = 1$ . The sequence is OEIS A192561 [11]; the indexing in (12.1) is maintained throughout. All logarithms of positive real quantities are natural logarithms, and  $W_0$  denotes the principal real Lambert function:  $W_0(x)e^{W_0(x)} = x$  for  $x > 0$ .

[write] **6 October 2026.** The live entry (A192561 revision #10, 25 June 2024; front matter) defines the sequence with the signed `Stirling1(n+1,k+1)^2`, which is (12.1); its 17 listed terms agree with the write's recomputation, and its term  $n = 10$  is (15.10). In this Part  $a = a_n$  is a saddle point and  $b_n$  the sequence; in Part I  $a_n$  is A192563 and  $b = \kappa_2$  (front matter, "Notation across the two Parts"). In the notation (18.3),  $b_n = \langle n!P_n, n!P_n \rangle_{\mathcal{F}}$ , while Part I's  $a_n = \langle n!P_n, \mathfrak{T}_n \rangle_{\mathcal{F}}$  with  $\mathfrak{T}_n(z) = \sum_k \left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} z^k$ : the two sequences pair the same polynomial with itself and with a second-kind polynomial (front matter, "One family, two methods").

The factorial weight is essential. It rewards configurations with many cycles and changes the analytic norm from a circle norm to a planar Gaussian norm. An asymptotic for the unweighted sum of squares cannot be substituted for (12.1).

### 12.1 An exact expansion before taking any limit

For  $n \geq 1$ , let  $a = a_n > 0$  solve

$$a = \sum_{j=1}^n \frac{1}{j+a}, \quad (12.2)$$

and let  $S_r = \sum_{j=1}^n (j+a)^{-r}$ . Write

$$e^{-at} \frac{P_n(a+t)}{P_n(a)} = \sum_{k \geq 0} c_k t^k, \quad C_{n,K} = \sum_{k=0}^{K-1} k! c_k^2. \quad (12.3)$$

The coefficients are real,  $c_0 = 1$ ,  $c_1 = 0$ , and the recurrence

$$k c_k = \sum_{r=2}^k (-1)^{r-1} S_r c_{k-r} \quad (k \geq 2) \quad (12.4)$$

uses only finite sums. We prove the exact statement

$$B_n = e^{-a^2} P_n(a)^2 (C_{n,K} + \varepsilon_{n,K}), \quad 0 \leq \varepsilon_{n,K} \leq (K+1) \left( \frac{K+1}{K} S_2 \right)^K \quad (12.5)$$

for every  $n \geq 1$  and  $K \geq 2$ . In particular, the error is at most  $e(K+1)\varphi^{-K}$ . This is an absolute bound on the normalized factor in parentheses; since that factor is at least 1, it also bounds the relative error made by retaining  $C_{n,K}$ .

A rational center replaces the algebraic root  $a$  when an exact arithmetic certificate is desired. The resulting recurrence has a nonzero linear term, and its explicit tail contains one additional positive exponential factor. Section 15 supplies both the theorem and a rational exponential enclosure.

## 12.2 All fixed logarithmic orders

Put  $w = W_0(en)$ . For every fixed integer  $R \geq 1$ , we prove

$$B_n = \frac{e^{w^2+1}}{2\pi w} \left\{ \sum_{r=0}^{R-1} \frac{A_r}{w^r} + O_R(w^{-R}) \right\}. \quad (12.6)$$

The first terms are

$$\frac{2\pi w B_n}{e^{w^2+1}} = 1 - \frac{1}{6w} + \frac{55}{72w^2} + \frac{1381}{6480w^3} + \frac{76541}{155520w^4} + O(w^{-5}). \quad (12.7)$$

The constant in the exponential is  $+1$ . Every coefficient is rational and generated symbolically, rather than fitted to data. No convergence of the series in  $1/w$ , and no growing-order error bound, is asserted.

## 12.3 Threshold inversion with rounding retained

Let  $N(T) = \min\{n \geq 0 : b_n \geq T\}$  for  $T > 1$ . With

$$L = \log T, \quad u = \frac{L}{2W_0(L/(2e))}, \quad D = \log u, \quad w_u = W_0(eu), \quad (12.8)$$

define  $Q_R(w) = \sum_{r=0}^{R-1} A_r w^{-r}$  and

$$v_R = u - \frac{w_u^2 + \log(u/w_u) + 1 + \log Q_R(w_u)}{2D}. \quad (12.9)$$

For each fixed  $R$ , there are constants  $C_R, T_R > 0$  such that, for  $T \geq T_R$ ,

$$\left\lceil v_R - C_R D^{-R-1} \right\rceil \leq N(T) \leq \left\lceil v_R + C_R D^{-R-1} \right\rceil. \quad (12.10)$$

The constants and onset are not numerically provided. Consequently, (12.10) is an asymptotic theorem, not a ready-to-use finite- $T$  numerical certificate. Actual certified threshold decisions can instead use the explicit Fock bounds and exact integer checks.

# 13 Combinatorial interpretation and Gaussian orthogonality

## 13.1 Matched cycles and the bivariate generating function

Consider a permutation on an  $n$ -element labeled set together with one special element. If the permutation has  $k+1$  cycles, exactly one cycle contains the special element. Thus  $\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}$  counts a distinguished cycle and  $k$  other cycles. Take two such permutations on separate labeled sets of size  $n$ , and a bijection between their  $k$  nondistinguished cycles. There are  $k!$  choices of this bijection. Summing gives (12.1). We use “matched cycles” as a descriptive name for this interpretation, without a historical naming claim.

Let  $L(x) = -\log(1-x)$ . A cycle containing the special element is an ordered list of ordinary labels, with exponential generating function  $(1-x)^{-1}$ . An ordinary cycle has generating function

$L(x)$ . A set of matched pairs of cycles on the two label types has generating function  $\exp(L(x)L(y))$ . Therefore

$$\frac{\exp(L(x)L(y))}{(1-x)(1-y)} = \sum_{m,n \geq 0} \left( \sum_{k \geq 0} \begin{bmatrix} m+1 \\ k+1 \end{bmatrix} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} k! \right) \frac{x^m y^n}{m!n!}. \quad (13.1)$$

In particular,  $B_n$  is its diagonal coefficient. Substituting  $x = y$  would sum all coefficients on an antidiagonal, not extract this diagonal. That distinction matters when comparing univariate alg-log results with the present problem.

**[write] 6 October 2026.** Equation (13.1) also follows termwise from  $\sum_m \begin{bmatrix} m+1 \\ k+1 \end{bmatrix} x^m/m! = L(x)^k/(k!(1-x))$ , the derivative of  $\sum_N \begin{bmatrix} N \\ k+1 \end{bmatrix} x^N/N! = L(x)^{k+1}/(k+1)!$ . Part I's double generating function (2.5) is the counterpart for the array A344639; there the diagonal is extracted by the one-variable coefficient formula (2.4), here by the norm (13.4).

## 13.2 The classical Fock identity

For an entire function  $f(t) = \sum_{k \geq 0} f_k t^k$ , define

$$\|f\|_{\mathcal{F}}^2 = \frac{1}{\pi} \int_{\mathbb{C}} e^{-|t|^2} |f(t)|^2 dA(t), \quad (13.2)$$

where  $A$  is planar Lebesgue area; the value is allowed to be infinite. Polar coordinates and circle orthogonality give

$$\|f\|_{\mathcal{F}}^2 = \sum_{k \geq 0} k! |f_k|^2. \quad (13.3)$$

Indeed, for each radius  $r$ , uniform convergence on the circle allows termwise integration and gives  $(2\pi)^{-1} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta = \sum_k |f_k|^2 r^{2k}$ . Tonelli's theorem then applies to the nonnegative sum, and  $2 \int_0^\infty e^{-r^2} r^{2k+1} dr = k!$ . Thus no exchange of a conditionally convergent series is involved. This is the standard Gaussian Fock norm [13].

The rising-factorial identity yields

$$P_n(z) = \frac{1}{n!} \sum_{k=0}^n \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} z^k, \quad B_n = \|P_n\|_{\mathcal{F}}^2. \quad (13.4)$$

This norm representation will be used directly; a two-variable contour deformation is unnecessary.

## 13.3 Translation and a global product bound

For any  $b \geq 0$ , set

$$H_b(t) = e^{-bt} \frac{P_n(b+t)}{P_n(b)}, \quad S_r(b) = \sum_{j=1}^n (j+b)^{-r}, \quad d = S_1(b) - b, \quad s = S_2(b). \quad (13.5)$$

Translating  $z = b+t$  in (13.2) cancels the linear terms from  $e^{-|z|^2}$  and  $e^{bt}$ , giving the exact identity

$$B_n = e^{-b^2} P_n(b)^2 \|H_b\|_{\mathcal{F}}^2. \quad (13.6)$$

The decisive estimate is global in  $t$ :

$$|H_b(t)|^2 \leq \exp\{2d \operatorname{Re} t + s|t|^2\}. \quad (13.7)$$

To prove it, write  $x_j = (j+b)^{-1} > 0$ . Then  $|1 + x_j t|^2 = 1 + 2x_j \operatorname{Re} t + x_j^2 |t|^2$ . The elementary inequality  $1 + u \leq e^u$  for real  $u$  applies, including when the left side vanishes. Multiply over  $j$  and include  $|e^{-bt}|^2$ . There is no logarithmic branch choice and no excluded zero in this argument.

## 14 The exact saddle and its uniform positive certificate

**Lemma 14.1** (The center and its moments). *For every  $n \geq 1$ , equation (12.2) has exactly one positive solution. At that solution,*

$$0 < S_2 \leq \min \left\{ \frac{a}{a+1}, \frac{1}{a} \right\} \leq \frac{1}{\varphi} < \frac{2}{3}. \quad (14.1)$$

Moreover, as  $n \rightarrow \infty$ ,

$$a \sim \log n, \quad S_r \leq \frac{a^{1-r}}{r-1} \quad (r \geq 2). \quad (14.2)$$

*Proof.* The function  $g(a) = \sum_{j=1}^n (j+a)^{-1} - a$  is continuous, strictly decreasing, positive at 0, and tends to  $-\infty$ . This proves existence and uniqueness. Since  $(j+a)^{-1} \leq (a+1)^{-1}$ ,  $S_2 \leq S_1/(a+1) = a/(a+1)$ . The decreasing-integral estimate gives

$$S_r \leq \int_0^n (a+x)^{-r} dx < \frac{a^{1-r}}{r-1}.$$

The functions  $a/(a+1)$  and  $1/a$  meet at  $a = \varphi$ , where both equal  $1/\varphi$ . One is increasing and the other decreasing, which proves (14.1).

Integral comparison in the defining equation also gives

$$\log \frac{a+n+1}{a+1} \leq a \leq \log \frac{a+n}{a}. \quad (14.3)$$

First  $a \rightarrow \infty$ , since a bounded  $a$  would make the sum in (12.2) diverge. Also  $a \leq H_n = O(\log n)$ . The left inequality in (14.3) is then  $a \geq \log n - O(\log \log n)$ , proving  $a/\log n \rightarrow 1$ .  $\square$

**Theorem 14.2** (Positive expansion and explicit tail). *Let  $n \geq 1$  and use (12.2)–(12.3). Then*

$$B_n = e^{-a^2} P_n(a)^2 C_n, \quad C_n = \sum_{k \geq 0} k! c_k^2 \geq 1, \quad (14.4)$$

where  $c_0 = 1$ ,  $c_1 = 0$ , and (12.4) holds. For every integer  $K \geq 1$  and every  $S_2 < q < 1$ ,

$$0 \leq C_n - C_{n,K} \leq \frac{(S_2/q)^K}{1-q}. \quad (14.5)$$

For fixed  $K$ , the right side is minimized at  $q = K/(K+1)$  whenever that point is admissible. It is admissible for every  $n \geq 1$  if  $K \geq 2$ , giving (12.5).

*Proof.* At  $b = a$ , the linear coefficient  $d$  in (13.5) is zero. Hence (13.7) becomes  $|H_a(t)|^2 \leq e^{S_2|t|^2}$ . By (14.1), this is integrable against the standard Gaussian. Equations (13.6) and (13.3) prove the positive expansion. Near  $t = 0$ ,

$$\log H_a(t) = \sum_{r \geq 2} \frac{(-1)^{r-1} S_r}{r} t^r \quad (|t| < a+1).$$

Differentiate this local identity and compare Taylor coefficients to obtain (12.4). Although the displayed logarithmic series is local,  $H_a$  itself is entire and the recurrence gives all its Taylor coefficients.

For the tail, apply (13.3) to  $H_a(\sqrt{q/S_2}t)$ . The global bound yields

$$\sum_{k \geq 0} k! c_k^2 \left( \frac{q}{S_2} \right)^k \leq \frac{1}{\pi} \int_{\mathbb{C}} e^{-(1-q)|t|^2} dA(t) = \frac{1}{1-q}.$$

Since  $q/S_2 > 1$ , deleting the first  $K$  terms and multiplying by  $(S_2/q)^K$  proves (14.5). Finally,

$$\frac{d}{dq} \log \frac{(S_2/q)^K}{1-q} = -\frac{K}{q} + \frac{1}{1-q}$$

changes sign from negative to positive at  $K/(K+1)$ . For  $K \geq 2$ , that point is at least  $2/3 > 1/\varphi \geq S_2$ . The estimate  $(1 + 1/K)^K \leq e$  finishes the stated simplification.  $\square$

A particularly simple choice works even at  $K = 1$ :  $q = 3/4$  gives

$$C_n - C_{n,K} \leq 4(4S_2/3)^K \leq 4 \left( \frac{4}{3\varphi} \right)^K. \quad (14.6)$$

Thus the exact expansion is uniformly convergent in this normalized sense over all positive indices. For a tolerance  $\epsilon > 0$ , choosing a  $K$  with the last expression at most  $\epsilon$  is a fully explicit universal rule. It is conservative: the actual  $S_2$  becomes small with  $n$ .

## 14.1 A broader negative real root theorem

**Theorem 14.3** (Finite products with negative zeros). *Let  $m \geq 1$ ,  $\lambda_j > 0$ , and  $P(z) = \prod_{j=1}^m (1 + z/\lambda_j)$ . Let  $a > 0$  be the unique solution of  $a = \sum_{j=1}^m (a + \lambda_j)^{-1}$ , and put  $S_r = \sum_{j=1}^m (a + \lambda_j)^{-r}$ . Then  $S_2 < 1$  and*

$$\|P\|_{\mathcal{F}}^2 = e^{-a^2} P(a)^2 \sum_{k \geq 0} k! c_k^2, \quad \sum_{k \geq K} k! c_k^2 \leq \frac{(S_2/q)^K}{1-q}$$

for  $S_2 < q < 1$ , where  $c_0 = 1$ ,  $c_1 = 0$ , and the recurrence (12.4) uses these moments.

*Proof.* Existence and uniqueness follow as before. Since  $a + \lambda_j > a$ ,

$$S_2 < \frac{1}{a} \sum_{j=1}^m \frac{1}{a + \lambda_j} = 1.$$

The translation, product inequality, coefficient recurrence, and scaled Gaussian argument are unchanged.  $\square$

Repeated roots are allowed. A nonzero constant multiplier is handled by multiplying the norm and prefactor by its squared modulus. The universal golden-ratio bound is specific to the arithmetic progression  $\lambda_j = j$ ; it is not asserted for arbitrary negative-root polynomials. The general certificate is an elementary consequence of classical Fock theory, and its scholarly priority in this formulation has not been established.

## 15 Rational centers and certified finite arithmetic

**Theorem 15.1** (Off-saddle certificate). *Let  $n \geq 1$ ,  $b \geq 0$ , and use (13.5). Suppose  $0 < s < q < 1$ , and write  $H_b(t) = \sum_{k \geq 0} c_k(b)t^k$ . Then  $c_0(b) = 1$ ,  $c_1(b) = d$ , and*

$$kc_k(b) = d c_{k-1}(b) + \sum_{r=2}^k (-1)^{r-1} S_r(b) c_{k-r}(b). \quad (15.1)$$

For every integer  $K \geq 1$ ,

$$B_n = e^{-b^2} P_n(b)^2 \left( \sum_{k < K} k! c_k(b)^2 + \varepsilon_{b,K} \right), \quad (15.2)$$

$$0 \leq \varepsilon_{b,K} \leq \frac{(s/q)^K}{1-q} \exp \left( \frac{qd^2}{s(1-q)} \right). \quad (15.3)$$

*Proof.* Translation proves (15.2); the local logarithmic derivative proves (15.1). Scale (13.7) by  $t \mapsto \sqrt{q/s} t$ . Writing  $t = x + iy$ , the Gaussian integral is

$$\frac{1}{\pi} \int_{\mathbb{R}^2} \exp\{-(1-q)(x^2 + y^2) + 2d\sqrt{q/s}x\} dx dy = \frac{1}{1-q} \exp \left( \frac{qd^2}{s(1-q)} \right).$$

Apply Fock orthogonality, remove the first  $K$  terms, and use  $(q/s)^k \geq (q/s)^K$  for  $k \geq K$ .  $\square$

### 15.1 A rational center with a guaranteed admissible scale

The root lies in  $[0, n]$ :  $g(0) > 0$  and  $g(n) = \sum_{j=1}^n (j+n)^{-1} - n < 0$ . Bisection evaluates  $g$  using rational arithmetic at every rational midpoint. After  $J$  steps, keep endpoints  $\ell \leq a \leq h$  with  $h - \ell = n2^{-J}$ , and choose  $b = h$ . Then

$$S_2(b) \leq S_2(a) \leq 1/\varphi, \quad d = g(b) \leq 0, \quad b - a \leq |d|. \quad (15.4)$$

The last inequality follows from  $g'(x) = -1 - S_2(x) \leq -1$ . For example,  $|d| \leq (1 + S_2(a))(b - a)$  by integration, so the extra exponential factor tends to 1 as the bracket narrows. No algebraic-number representation of  $a$  is needed.

The scales  $q = 3/4$  and, for  $K \geq 2$ ,  $q = K/(K+1)$  remain admissible. When  $d \neq 0$ , the latter is *not generally the minimizer* of (15.3): the extra exponential also depends on  $q$ . Calling it the optimal off-saddle scale would be incorrect. It is a convenient explicit admissible choice.

### 15.2 Rational upper and lower bounds for the exponential

For rational  $x \geq 0$  and integer  $M \geq 0$  with  $x < M + 2$ , put

$$E_M^-(x) = \sum_{j=0}^M \frac{x^j}{j!}, \quad E_M^+(x) = E_M^-(x) + \frac{x^{M+1}}{(M+1)!} \frac{1}{1 - x/(M+2)}. \quad (15.5)$$

Every quantity is rational, and

$$E_M^-(x) \leq e^x \leq E_M^+(x). \quad (15.6)$$



## 16 Exact center asymptotics and the Gaussian correction

The explicit tail already contains an all-orders exact-center expansion. For each fixed  $K$ , take  $q = 3/4$  in (14.5) and use  $S_2 = O(a^{-1})$  to obtain

$$B_n = e^{-a^2} P_n(a)^2 \{C_{n,K} + O_K(a^{-K})\}. \quad (16.1)$$

The coefficients  $c_k$  in this statement are the exact finite- $n$  moment polynomials. Replacing moments by their leading values without controlling the accumulated errors would change the claim.

The exponential formula gives a second finite description:

$$c_k = \sum_{\substack{m_2, \dots, m_k \geq 0 \\ \sum_{r=2}^k r m_r = k}} \prod_{r=2}^k \frac{1}{m_r!} \left( \frac{(-1)^{r-1} S_r}{r} \right)^{m_r}. \quad (16.2)$$

Assign weight  $r-1$  to  $S_r$  because  $S_r = O(a^{1-r})$ . In a monomial of  $c_k$ , the weight is  $k - \sum m_r \geq \lceil k/2 \rceil$ . Hence the product of two monomials from  $c_k$  has weight at least  $2 \lceil k/2 \rceil \geq k$ . Only finitely many  $k$  can contribute below any fixed weight. This observation, together with the tail bound, rigorously justifies collecting the expansion by weight.

The first coefficients are

$$\begin{aligned} c_2 &= -S_2/2, & c_3 &= S_3/3, & c_4 &= S_2^2/8 - S_4/4, \\ c_5 &= S_5/5 - S_2 S_3/6, & c_6 &= -S_6/6 + S_2 S_4/8 + S_3^2/18 - S_2^3/48. \end{aligned}$$

Squaring with the  $k!$  weights and retaining total weight at most six gives

$$\begin{aligned} C_n &= 1 + \frac{1}{2} S_2^2 + \frac{3}{8} S_2^4 + \frac{2}{3} S_3^2 - \frac{3}{2} S_2^2 S_4 \\ &\quad + \frac{5}{16} S_2^6 + \frac{10}{3} S_2^2 S_3^2 + \frac{3}{2} S_4^2 + O(a^{-7}). \end{aligned} \quad (16.3)$$

For example,  $24c_4^2 = 3S_2^4/8 - 3S_2^2 S_4/2 + 3S_4^2/2$ . The  $c_5^2$  term contributes  $10S_2^2 S_3^2/3$  at weight six, and the  $c_6^2$  term contributes  $5S_2^6/16$  there.

[write] **6 October 2026.** Checked by the write symbolically (SymPy):  $c_2, \dots, c_6$  above and every term of (16.3).

Keeping just the quadratic part of  $\log H_a$  gives  $e^{-S_2 t^2/2}$ . Its norm is exactly

$$\left\| e^{-S_2 t^2/2} \right\|_{\mathcal{F}}^2 = \sum_{j \geq 0} \frac{(2j)!}{(j!)^2} \left( \frac{S_2^2}{4} \right)^j = \frac{1}{\sqrt{1 - S_2^2}}. \quad (16.4)$$

Thus the two-dimensional Gaussian approximation is

$$B_n = \frac{e^{-a^2} P_n(a)^2}{\sqrt{1 - S_2^2}} \{1 + O(a^{-4})\}. \quad (16.5)$$

Its first nonquadratic contribution in the normalized norm is  $2S_3^2/3$ , of order  $a^{-4}$ . The complete finite positive sum, with (14.5), is preferable when a numerical certificate is required; (16.3) is an explanatory asymptotic rearrangement.

## 17 A universal Gamma norm at the Lambert anchor

The exact center is ideal for finite- $n$  certification. To express coefficients as rational numbers independent of  $n$ , use the Lambert anchor

$$w = W_0(en), \quad b_* = w - 1, \quad \log n = w + \log w - 1. \quad (17.1)$$

For  $w \geq 2$ , define the entire function and its norm

$$h_w(t) = \frac{w^t \Gamma(w)}{\Gamma(w+t)}, \quad \mathcal{H}(w) = \|h_w\|_{\mathcal{F}}^2. \quad (17.2)$$

Here  $w^t = e^{t \log w}$ . We will use

$$d_w = \log w - \psi(w), \quad s_w = \zeta(2, w) = \sum_{j \geq 0} (w+j)^{-2}, \quad (17.3)$$

where  $\psi = \Gamma'/\Gamma$ .

The reciprocal-Gamma canonical product gives

$$h_w(t) = e^{d_w t} \prod_{j \geq 0} \left(1 + \frac{t}{w+j}\right) e^{-t/(w+j)}. \quad (17.4)$$

The product converges locally uniformly. Applying the same factor inequality as before to its finite partial products and passing to the limit yields

$$|h_w(t)|^2 \leq e^{2d_w \operatorname{Re} t + s_w |t|^2}. \quad (17.5)$$

As  $w \rightarrow \infty$ ,  $d_w = O(w^{-1})$  and  $s_w = O(w^{-1})$ . In particular the norm is finite for all sufficiently large  $w$ , and  $\mathcal{H}(w) \geq 1$  by the constant coefficient in (13.3). In fact  $s_w < 1$  for  $w \geq 2$ .

**Proposition 17.1** (Reduction to the universal norm). *As  $n \rightarrow \infty$ , with  $w = W_0(en)$ ,*

$$B_n = \frac{n^{2(w-1)} e^{-(w-1)^2}}{\Gamma(w)^2} \mathcal{H}(w) \{1 + O(w^2/n)\}. \quad (17.6)$$

*Proof.* Let  $h_{n,w}(t) = e^{-(w-1)t} P_n(w-1+t)/P_n(w-1)$ . Its linear logarithmic coefficient and quadratic majorant are

$$\begin{aligned} d_{n,w} &= \psi(n+w) - \psi(w) - (w-1) = O(w^{-1}) + O(w/n), \\ s_{n,w} &= \sum_{j=0}^{n-1} (w+j)^{-2} = O(w^{-1}). \end{aligned}$$

The global estimate (13.7) therefore shows that both  $e^{-|t|^2} |h_{n,w}(t)|^2$  and  $e^{-|t|^2} |h_w(t)|^2$  are bounded by  $Ce^{-|t|^2/2}$  for all sufficiently large  $n$ , with a fixed  $C$ . This follows, for example, by absorbing the linear term into a small multiple of  $|t|^2$  and a constant.

The Gamma product identity gives

$$h_{n,w}(t) = h_w(t) R_{n,w}(t), \quad R_{n,w}(t) = \frac{\Gamma(n+w+t)}{\Gamma(n+w)n^t}. \quad (17.7)$$

On  $|t| \leq \sqrt{n}/4$ , the arguments of the Gamma function in  $R_{n,w}$  lie in the positive half-plane. Thus  $R_{n,w}$  has a consistent analytic logarithm there. Identity (17.7) is interpreted by analytic continuation at zeros of  $h_w$ , so division by a vanishing function is not needed. The standard uniform digamma estimate in that half-plane gives

$$\psi(n+w+u) - \log n = O\left(\frac{w+|u|+1}{n}\right) \quad (|u| \leq \sqrt{n}/4).$$

Integrating along the segment from 0 to  $t$  shows that

$$\log R_{n,w}(t) = O\left(\frac{(w+1)|t| + |t|^2}{n}\right). \quad (17.8)$$

The digamma estimate follows from its usual sectorial asymptotic expansion [14], here used far from the negative axis.

Use  $|e^v - 1| \leq |v|e^{|v|}$  in (17.7). The exponential factor from (17.8) can be absorbed into the global Gaussian majorant, since its quadratic coefficient is  $O(1/n)$  and its linear coefficient is  $O(w/n)$ . Gaussian integrals of  $|t|$  and  $|t|^2$  then imply a norm difference  $O(w/n)$  on this disk. Outside the disk, both global bounds give tails  $O(e^{-cn})$  for some  $c > 0$ . Consequently,

$$\|h_{n,w}\|_{\mathcal{F}}^2 = \mathcal{H}(w) + O(w/n) = \mathcal{H}(w)\{1 + O(w/n)\}. \quad (17.9)$$

The last step uses  $\mathcal{H}(w) \geq 1$ .

Finally,  $P_n(w-1) = \Gamma(n+w)/(\Gamma(n+1)\Gamma(w))$  and

$$\frac{\Gamma(n+w)}{\Gamma(n+1)n^{w-1}} = 1 + O(w^2/n).$$

This follows by integrating the real digamma estimate between  $n+1$  and  $n+w$ , an interval of length  $O(w)$ . Combine this prefactor estimate, translation, and (17.9).  $\square$

The error  $w^2/n$  is smaller than every fixed power  $w^{-1}$ , since  $n = we^{w-1}$ . This observation permits an all-orders inverse-logarithm expansion, but does not identify a complete exponentially small correction or transseries.

**[write] 6 October 2026.** At practical sizes the  $O(w^2/n)$  term of (17.6) is not small against the inverse-logarithm terms. With exact  $b_n$ , the intake found  $b_n$  divided by the leading term of (18.11) to be 1.0972 at  $n = 200$  ( $w = 4.742$ ) and 1.0265 at  $n = 1000$  ( $w = 6.100$ ), and the scaled residual  $(2\pi w b_n / ((n!)^2 e^{w^2+1}) - Q_2(w))w^2$  to be 2.976 and 2.004, against its limit  $A_2 = 55/72 = 0.764$ ; the write recomputed the values at  $n = 200$ , and 2.530 at  $n = 400$ . The excess over  $A_2$ , namely 2.21, 1.77 and 1.24 at  $n = 200, 400, 1000$ , is 0.87, 0.88 and 0.90 times  $w^4/n = 2.53, 2.00, 1.38$ , i.e. of the size of the reduction error  $O(w^2/n)$  multiplied by  $w^2$ . This is consistent with Theorem 18.2, under which the scaled residual tends to  $A_2$  only as  $n \rightarrow \infty$ ; the manuscript does not compute the  $1/n$ -scale correction (Subsection 22.3).

## 18 All orders by a finite coefficient argument

The majorant (17.5) makes the all-orders proof particularly short: truncate the positive norm first, and only then expand the finitely many retained coefficients. There is no need to expand the integrand on a growing complex disk.

## 18.1 The coefficient generator

Use the Bernoulli convention  $B_1(t) = t - 1/2$ , and set

$$g_m(t) = \frac{(-1)^m}{m(m+1)} \{B_{m+1}(t) - B_{m+1}(0)\} \quad (m \geq 1). \quad (18.1)$$

Define real polynomials  $p_r(t)$  by

$$p_0(t) = 1, \quad p_r(t) = \frac{1}{r} \sum_{m=1}^r m g_m(t) p_{r-m}(t). \quad (18.2)$$

Equivalently,  $\sum_{r \geq 0} p_r(t) x^r = \exp(\sum_{m \geq 1} g_m(t) x^m)$  as a formal series. Define the finite polynomial inner product

$$\langle f, g \rangle_{\mathcal{F}} = \sum_{k \geq 0} k! [t^k] f(t) [t^k] g(t), \quad H_r = \sum_{j=0}^r \langle p_j, p_{r-j} \rangle_{\mathcal{F}}. \quad (18.3)$$

No complex conjugation is needed here because all coefficients are real.

The shifted Gamma expansion [14, Eq. 5.11.8] gives, on each fixed  $t$ -disk,

$$\log h_w(t) \sim \sum_{m \geq 1} g_m(t) w^{-m}. \quad (18.4)$$

For example,

$$g_1(t) = -\frac{t^2}{2} + \frac{t}{2}, \quad g_2(t) = \frac{t^3}{6} - \frac{t^2}{4} + \frac{t}{12},$$

and

$$p_1(t) = g_1(t), \quad p_2(t) = \frac{t^4}{8} - \frac{t^3}{12} - \frac{t^2}{8} + \frac{t}{12}.$$

The nonzero linear term in  $p_1$  reflects the fact that  $w - 1$  is an anchor rather than the exact saddle.

**Theorem 18.1** (Universal norm expansion). *For every fixed integer  $R \geq 1$ ,*

$$\mathcal{H}(w) = \sum_{r=0}^{R-1} \frac{H_r}{w^r} + O_R(w^{-R}) \quad (w \rightarrow \infty), \quad (18.5)$$

where (18.1)–(18.3) give the coefficients.

*Proof.* Write  $h_w(t) = \sum_{k \geq 0} h_k(w) t^k$ . The scaled-Fock argument used in Theorem 15.1, now applied to (17.5), gives

$$\sum_{k \geq K} k! h_k(w)^2 \leq \frac{(s_w/q)^K}{1-q} \exp\left(\frac{q d_w^2}{s_w(1-q)}\right) \quad (s_w < q < 1). \quad (18.6)$$

For fixed  $K$  and all sufficiently large  $w$ , take  $q = 1/2$ . Since  $s_w = O(w^{-1})$  and  $d_w = O(w^{-1})$ , while  $s_w \geq \int_0^\infty (w+x)^{-2} dx = 1/w$ , the exponent is  $O(w^{-1})$ . The right side is therefore  $O_K(w^{-K})$ .

Choose  $K = R$ . The norm differs from the finite sum  $\sum_{k < R} k! h_k(w)^2$  by  $O_R(w^{-R})$ . Each fixed  $h_k$  is determined by

$$k h_k = d_w h_{k-1} + \sum_{j=2}^k (-1)^{j-1} \zeta(j, w) h_{k-j}, \quad h_0 = 1. \quad (18.7)$$

The fixed-order digamma and Hurwitz-zeta expansions yield a rigorous finite asymptotic expansion of every such coefficient. Equivalently, Cauchy's coefficient formula on one fixed disk applied to the uniformly valid fixed-disk expansion (18.4) gives

$$h_k(w) = \sum_{r=0}^{R-1} [t^k] p_r(t) w^{-r} + O_{R,k}(w^{-R}).$$

Only finitely many coefficients occur, so multiplication and summation preserve an  $O_R(w^{-R})$  remainder.

It remains to check that truncating  $k < R$  does not discard any coefficient  $H_r$  for  $r < R$ . From (18.2),  $\deg p_r \leq 2r$ , and  $p_r(0) = 0$  for  $r > 0$ . If  $a + b = r$  and a monomial degree  $k$  occurs in both  $p_a$  and  $p_b$ , then  $k \leq \min(2a, 2b) \leq r < R$ ; the cases  $a = 0$  or  $b = 0$  obey the same bound. Thus all the terms defining  $H_0, \dots, H_{R-1}$  survive the finite norm truncation. Together with (18.6), this proves (18.5).  $\square$

## 18.2 Stirling normalization and the constant term

Define a formal series  $E(x) = \sum_{r \geq 0} E_r x^r$  by

$$E(x) = \exp \left( - \sum_{j \geq 1} \frac{2B_{2j}}{2j(2j-1)} x^{2j-1} \right), \quad A_r = \sum_{j=0}^r E_j H_{r-j}. \quad (18.8)$$

Here  $B_{2j} = B_{2j}(0)$ . Stirling's expansion and  $\log n = w + \log w - 1$  show that

$$\begin{aligned} \log \frac{n^{2(w-1)} e^{-(w-1)^2}}{\Gamma(w)^2} &= w^2 + 1 - \log(2\pi w) \\ &\quad - \sum_{j \geq 1} \frac{2B_{2j}}{2j(2j-1)w^{2j-1}}, \end{aligned} \quad (18.9)$$

where the sum is interpreted asymptotically to each fixed order. For clarity, the elementary part of the cancellation is

$$2(w-1)(w + \log w - 1) - (w-1)^2 - 2(w-1/2) \log w + 2w = w^2 + 1 - \log w.$$

This fixes both the  $+1$  and the factor  $w^{-1}$ .

**Theorem 18.2** (Complete inverse logarithm expansion). *With  $A_r$  defined by (18.1)–(18.8), for each fixed integer  $R \geq 1$ ,*

$$b_n = \frac{(n!)^2 e^{W_0(en)^2+1}}{2\pi W_0(en)} \left( \sum_{r=0}^{R-1} \frac{A_r}{W_0(en)^r} + O_R(W_0(en)^{-R}) \right). \quad (18.10)$$

*In particular,*

$$b_n \sim \frac{(n!)^2 e^{W_0(en)^2+1}}{2\pi W_0(en)}. \quad (18.11)$$

*Proof.* Multiply Proposition 17.1, Theorem 18.1, and the fixed-order exponential of (18.9). The relative error  $O(w^2/n)$  is  $o(w^{-R})$  for every fixed  $R$ , and the remaining products are finite asymptotic algebra. Their coefficients are exactly the convolution (18.8).  $\square$

### 18.3 Coefficients and an exact generation procedure

The first eleven values are shown in Table 2. The companion independently implements rational polynomial addition, multiplication, the Bernoulli recurrence, and (18.2)–(18.8). No symbolic algebra dependency or numeric fitting is required for these coefficients.

Table 2: Universal norm and normalized sequence coefficients.

| $r$ | $H_r$                 | $A_r$                                  |
|-----|-----------------------|----------------------------------------|
| 0   | 1                     | 1                                      |
| 1   | 0                     | $-1/6$                                 |
| 2   | $3/4$                 | $55/72$                                |
| 3   | $1/3$                 | $1381/6480$                            |
| 4   | $155/288$             | $76541/155520$                         |
| 5   | $259/720$             | $1807553/6531840$                      |
| 6   | $2183/5760$           | $385303949/1175731200$                 |
| 7   | $2809/60480$          | $-66416099/7054387200$                 |
| 8   | $3129727/9676800$     | $108926247893/338610585600$            |
| 9   | $-3827969/7257600$    | $-584017596861539/1005673439232000$    |
| 10  | $138901163/116121600$ | $108854736210330631/84476568895488000$ |

For a check at the first nontrivial order,  $H_1 = 2\langle 1, p_1 \rangle = 0$ , while  $H_2 = \langle p_1, p_1 \rangle = 1/4 + 2!/4 = 3/4$ . The first Stirling coefficients are  $E_0 = 1$ ,  $E_1 = -1/6$ ,  $E_2 = 1/72$ , and  $E_3 = 31/6480$ . Consequently  $A_2 = 3/4 + 1/72 = 55/72$  and  $A_3 = 1/3 - (1/6)(3/4) + 31/6480 = 1381/6480$ .

For general  $R$ , if  $\eta_m$  denotes the coefficient of  $x^m$  in the exponent of (18.8), the recurrence

$$E_0 = 1, \quad E_r = \frac{1}{r} \sum_{m=1}^r m \eta_m E_{r-m} \quad (18.12)$$

is sufficient. Thus all operations at any requested fixed order are exact rational operations on finitely many polynomials. The existence of this generator should not be confused with a uniform estimate as the order itself grows with  $n$ .

## 19 Inverses and exact integer thresholds

### 19.1 Strict increase and the leading inverse

The sequence is strictly increasing. One direct proof uses (12.1) and the unsigned Stirling recurrence

$$\begin{bmatrix} m+1 \\ j \end{bmatrix} = m \begin{bmatrix} m \\ j \end{bmatrix} + \begin{bmatrix} m \\ j-1 \end{bmatrix}. \quad (19.1)$$

Alternatively,  $P_{n+1}(z) = P_n(z)(1 + z/(n+1))$  has nonnegative coefficients strictly larger in at least one degree, so its squared Fock norm strictly increases;  $(n!)^2$  also increases except at the initial harmless equality  $0! = 1!$ . In particular  $b_1 = 2 > b_0$ . Since  $b_n \geq (n!)^2$ , the sequence is unbounded, and  $N(T)$  is well defined for  $T > 1$ .

By Stirling's formula and Theorem 18.2,

$$\log b_n = 2n(\log n - 1) + O((\log n)^2). \quad (19.2)$$

The equation  $2u(\log u - 1) = L$  has the exact positive solution

$$u = \frac{L}{2W_0(L/(2e))}, \quad L = \log T. \quad (19.3)$$

Monotonicity and (19.2) show that  $N(T) \sim u$ . The omitted logarithmic correction induces an additive shift of order  $\log u$ , so the leading inverse alone is not an integer threshold rule.

## 19.2 The smooth model and an implicit two-sided bracket

For fixed  $R \geq 1$ , let

$$F_R(x) = 2 \log \Gamma(x+1) + w(x)^2 + 1 - \log(2\pi w(x)) + \log Q_R(w(x)), \quad w(x) = W_0(ex). \quad (19.4)$$

For all sufficiently large real  $x$ ,  $Q_R(w(x)) > 0$ , so this is smooth. Since  $w'(x) = w(x)/(x(1+w(x)))$ ,

$$F'_R(x) \sim 2 \log x > 0, \quad F''_R(x) = O(1/x). \quad (19.5)$$

The forward expansion gives

$$\log b_n = F_R(n) + O_R(w(n)^{-R}). \quad (19.6)$$

The logarithm is legitimate because  $Q_R(w) \rightarrow 1$ .

Choose a sufficiently large fixed constant  $D_R$  so that the absolute remainder in (19.6) is at most  $D_R w(n)^{-R}$  for all sufficiently large  $n$ . For large  $T$ , let  $x_-$  and  $x_+$  be the large roots of

$$F_R(x_-) + D_R w(x_-)^{-R} = \log T, \quad F_R(x_+) - D_R w(x_+)^{-R} = \log T. \quad (19.7)$$

The perturbed model functions are increasing in the relevant range, by (19.5). If an integer  $n < x_-$ , then its upper logarithmic bound is strictly below  $\log T$ , so  $b_n < T$ . If  $n \geq x_+$ , its lower bound is at least  $\log T$ , so  $b_n \geq T$ . This proves

$$\lfloor x_- \rfloor \leq N(T) \leq \lceil x_+ \rceil, \quad x_+ - x_- = O_R((\log u)^{-R-1}). \quad (19.8)$$

For sufficiently large  $T$ , every index below the asymptotic onset is also excluded, because the finitely many corresponding  $b_n$  have a finite maximum. The equality case at the threshold is preserved by the upper bound's weak inequality. Replacing the lower ceiling by a floor would give a valid but weaker bound; automatic rounding of a single real root is not justified.

## 19.3 An explicit nested Lambert center to all fixed orders

**Theorem 19.1** (Explicit inverse center). *Let  $u, D, w_u$  be as in (12.8), and let  $v_R$  be as in (12.9). For every fixed  $R \geq 1$ , there exist  $C_R, T_R > 0$  such that (12.10) holds for  $T \geq T_R$ . The unrounded real interval has width  $O_R(D^{-R-1})$ ; its two ceilings may still differ by one.*

*Proof.* Put

$$J_R(u) = w_u^2 + \log(u/w_u) + 1 + \log Q_R(w_u). \quad (19.9)$$

Stirling's expansion of  $2 \log \Gamma(u + 1)$  gives

$$F_R(u) - L = J_R(u) + \frac{1}{6u} + O(u^{-3}). \quad (19.10)$$

Here  $w_u = O(D)$ ,  $J_R(u) = O(D^2)$ , and  $J'_R(u) = O(D/u)$ . The definition  $v_R = u - J_R(u)/(2D)$  therefore has  $v_R - u = O(D)$ . Uniformly on this  $O(D)$  neighborhood of  $u$ , direct differentiation of (19.4) yields

$$F'_R(x) = 2D + O(D/u), \quad F''_R(x) = O(1/u).$$

Taylor's theorem and (19.10) then show that

$$F_R(v_R) - L = O(D^2/u). \quad (19.11)$$

Indeed the leading term  $J_R(u)$  is canceled exactly by  $2D(v_R - u)$ , while the derivative correction times the shift and the quadratic remainder are both  $O(D^2/u)$ .

The smooth root  $y_R$  of  $F_R(y_R) = L$  consequently satisfies  $y_R - v_R = O(D/u)$ . Because  $D/u$  is smaller than every fixed power  $D^{-1}$ , this error is negligible relative to  $D^{-R-1}$ . The perturbed roots in (19.7) each differ from  $y_R$  by  $O_R(D^{-R-1})$ : their perturbation is  $O_R(D^{-R})$  and their derivative is of order  $D$ . Enclose both roots in  $[v_R - C_R D^{-R-1}, v_R + C_R D^{-R-1}]$  and apply (19.8).  $\square$

This expression uses two principal Lambert evaluations and the finite coefficient polynomial  $Q_R$ . Its shift-Taylor error is smaller than every inverse-logarithm order. That statement does not supply an exponentially improved inverse, nor a full transseries. The constants and a numerical starting threshold for the asymptotic windows have not been derived in this report; there is no claim that they are intrinsically noncomputable.

[write] **6 October 2026.** Relation to the transseries volume [19] (front matter): (19.3) is an instance of `p0:prop:factorial-core` with  $\kappa = 2$ ,  $d = -2$ , which gives  $v = W_0(L/(2e))$  and  $X = e^{v+1} = L/(2v) = u$ , the unique solution with  $2 \log u - 2 > 0$ . The centre  $v_R$  and the brackets (19.8) and (12.10) are proved here from the smooth model  $F_R$ , which is not an interpolation of  $(b_n)$ ; they are not instances of `p0:thm:staircase` and play the role of its separation condition (2).

## 19.4 A practical certificate protocol

For a finite threshold  $T$ , the exact interval (15.8) supports direct decisions:

1. At a candidate index  $n$ , compute rational  $\mathcal{L}_n \leq b_n \leq \mathcal{U}_n$ . If  $\mathcal{U}_n < T$ , then  $n < N(T)$ ; if  $\mathcal{L}_n \geq T$ , then  $n \geq N(T)$ .
2. If the interval straddles  $T$ , increase the Fock truncation and the exponential accuracy, and if useful refine the rational center. Do not convert an inconclusive comparison into a decision.
3. When equality could be present, use the integer isolation criterion (15.9) or the exact recurrence (19.1). Convergence to a boundary from both sides alone need not decide equality.
4. Use monotonicity to bracket the index, then test adjacent integers. A certificate  $b_{m-1} < T \leq b_m$  proves  $N(T) = m$ .

All comparisons with an exact rational or integer threshold can remain rational. For a threshold specified only approximately, its own uncertainty must also be enclosed. The asymptotic center is useful for choosing a candidate, while the finite arithmetic certificate supplies the actual decision.

## 20 Sources and relation to earlier methods

### 20.1 What the source inspection establishes

The inspected OEIS entry and official data mirror give the defining factorial-weighted sum and finite terms [11]; no asymptotic was present in the inspected snapshot. Focused searches covered the identifier, number prefix, factorial-weighted Stirling squares, matched cycles, rising-factorial Fock norms, and relevant multivariate or alg-log methods. No source containing this sequence-specific equivalent and coefficient series was located. This is a bounded search result, not proof of novelty.

Wilf [12] proves that the probability that two independent uniform permutations on  $n$  letters have the same cycle count is asymptotic to  $1/(2\sqrt{\pi \log n})$ . His proof turns the unweighted squared-Stirling sum into a circle norm and applies Laplace’s method. This is a close conceptual precedent. The  $k!$  weight in the present sequence leads instead to the standard planar Gaussian norm. No new general norm identity or saddle method is being claimed.

Alpay and Porat [13] discuss generalized Fock spaces whose coefficient norms involve factorial powers; their classical case includes (13.3). Their work concerns a broader functional-analytic setting and Stirling-number operators. The identity used here is classical, regardless of whether the sequence-specific evaluation appears in that paper.

The shifted Gamma and digamma expansions used in Sections 17 and 18 are standard; DLMF equations 5.11.1, 5.11.2, and 5.11.8 are direct references [14]. The proof added here organizes these inputs around an exact positive expansion and a scaled tail, so that the all-orders norm remainder involves finitely many coefficient expansions.

### 20.2 General asymptotic frameworks and remaining overlap

Gittenberger and Mandlbürger [15] develop multivariate Hayman admissibility and Gaussian coefficient asymptotics. Their cited admissibility setup uses entire functions, whereas (13.1) has logarithmic singularities. That particular theorem is not directly applicable without additional work. The general idea of a multivariate Gaussian coefficient estimate is established prior art.

Bender and Richmond [16] give a general framework for labelled structures by number of components. The official abstract and indexed primary theorem material were inspected, but a full PDF reading was not completed in the source pass. Their theorem may provide an alternative route to a leading saddle estimate once its hypotheses are checked; this report does not claim otherwise. It also does not infer that the positive certificate or the specific coefficients are absent from every related paper.

Gao, Laferrière, and Panario [17] treat a univariate  $\exp(C(z))$  schema of alg-log type with positive logarithmic exponent. The official abstract was inspected; the full PDF could not be read through the available reader in the source pass. The substitution  $x = y$  in (13.1) gives an antidiagonal sum and cannot by itself identify  $b_n$ . Detailed overlap beyond that schema distinction remains unverified.

Accordingly, the defensible contribution of this report is a self-contained sequence-specific derivation, its explicit all-orders rational coefficients and inverse, and an exact finite- $n$  certification procedure. A first-ever claim for either the general certificate or the leading method would require a broader literature investigation.

**[write] 6 October 2026.** The write did not read Wilf’s note, Alpay–Porat, Gittenberger–Mandlbürger, Bender–Richmond or Gao–Laferrière–Panario; what is said here about them, and the inspection levels in

`data/155-matched-SOURCE_PROVENANCE.json`, are the manuscript's. No other report of the repository treats A192561 (repository search, 6 October 2026).

## 21 Computation and reproducibility

The source archive contains this report, its editable LaTeX, a Python standard-library companion, fixed inputs and reproducible results, source provenance, and deterministic build and packaging scripts. It contains no earlier report, unrelated data, or private review notes.

### 21.1 Three different kinds of evidence

1. **Proof.** Theorems 14.2, 15.1, 18.1, 18.2, and 19.1 establish their stated mathematical ranges. Finite experiments do not supply the all-orders argument.
2. **Exact reproducible computation.** Rational polynomial arithmetic generates the displayed coefficients. Integer Stirling arithmetic evaluates finite terms. The rational-center certificate uses exact fractions and (15.5); its interval and unique-integer decision are rigorous for the actual chosen finite parameters.
3. **Optional numerical diagnostics.** High-precision floating-point comparisons can corroborate the positive sums and inspect normalized residuals of the Gamma norm. They cannot turn floating-point evaluations into interval proofs. The source archive prioritizes the exact route and has no numerical-library dependency.

### 21.2 Companion design and safe output behavior

The companion exposes bounded commands for its fixed certificate and coefficient checks. Numeric input validation uses explicit exceptions, so Python's optimized mode does not disable it. Imports do not write output. Generated files use new exclusive paths, reject parent traversal and symlink paths, and never silently overwrite an existing result. Exact fractions are serialized without converting proof decisions to floating point. The companion README gives the precise commands and resource bounds; its checked results can be recomputed using ordinary Python and again with `python3 -O`.

The PDF builder assumes a trusted parent directory with no concurrent adversarial path replacement. It creates an exclusive clean directory, disables shell escape, pins deterministic metadata, and repeats TeX until references stabilize. The archive builder verifies a fixed file allowlist against `SHA256SUMS`, uses fixed ZIP metadata, and refuses an existing output destination. Static symlinks and parent traversal are rejected; the directory builder is not claimed to resist malicious replacement between directory creation and opening. The manifest covers every payload file; it excludes itself to avoid a self-hash cycle. The ZIP includes that manifest. Rebuilding twice in the same recorded toolchain is required to obtain byte-identical PDFs and archives. Cross-version TeX byte identity is not asserted.

[write] 6 October 2026. Shipped names: `code/155-matched-companion-fock_exact.py`, `-safe_io.py` and `-test_companion.py`, the inputs and results in `data/155-matched-companion-*`, the companion README as `155-matched-companion-README.md`, the release scripts as `code/155-matched-*.py`. The companion's own `SHA256SUMS`, the root manifest, `Report155.pdf` and the delivery README are not shipped. At the write (Python 3.14.4, Windows, fresh extraction) 24 of the 28 tests passed; the four `SafeIO` tests stop

because they need POSIX `O_NOFOLLOW/O_DIRECTORY` or `os.mkfifo`, as the package requires; all mathematical tests passed. The delivered `verification.json` records 28 of 28 on Linux.

## 22 Further questions and limitations

### 22.1 Root classes beyond the arithmetic progression

Theorem 14.3 allows arbitrary positive root parameters, but only gives the instance-specific condition  $S_2 < 1$ . For regularly spaced roots, roots with a limiting density, or families with a quantitative gap from zero, one can ask for sharp uniform bounds on  $S_2$  and explicit rates of decay as the degree grows. Such hypotheses may produce other universal positive certificates and asymptotic scales. Polynomials with nonreal or positive zeros need separate arguments; the present product estimate alone does not provide the same real saddle or normalized tail.

### 22.2 Computational complexity and interval algorithms

The demonstration uses exact fractions because they make every comparison transparent. Their numerators and denominators can become large. Useful next steps include complexity estimates in  $n$ ,  $K$ , and requested accuracy; rigorous ball arithmetic; range reduction for exponential enclosures; and adaptive choices of the center and scale. For an off-saddle center, minimizing (15.3) rather than reusing  $K/(K+1)$  may substantially improve bounds. An efficiency comparison with integer Stirling recurrence or fast polynomial methods would require a dedicated benchmark and is not settled by the example.

### 22.3 Beyond fixed logarithmic orders

The proof separates an  $O(w^2/n)$  finite-product correction from a complete fixed-order expansion in  $w^{-1}$ . It does not compute the first correction on the  $1/n$  scale, establish optimal truncation, control coefficients at growing order, or prove Borel summability. A systematic two-scale expansion could improve both the forward formula and its inverse. Quantifying explicit constants and starting indices for the logarithmic remainders would also turn the asymptotic inverse windows into independent numerical tools. Those tasks are distinct from the already explicit finite- $n$  Fock certificate.

### 22.4 Other factorial weights and statistics

Generalized Fock norms suggest other factorial weights, such as  $\sum_k \left[ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right]^2 (k!)^\alpha$ , which may require different saddle geometry. Only the weight  $k!$  and the negative-root polynomial extension are proved here. Weighted cycle-count distributions, local limits, and concentration remain separate research questions.

[write] **6 October 2026.** Section 23 collects these questions with those of Part I under Vladimir’s standing rule.

## 23 Further questions and research

[write] *Added in the write, 6 October 2026.* This section collects the open questions of the two Parts (Sections 11 and 22) and their stated non-claims under Vladimir's standing rule of 4 October 2026: each is stated with its source, the sketch the source gives, and what is missing. Neither manuscript contains a claim found to be wrong. The third member of the family, A192562, is treated by neither manuscript; the write proves an exact identity for it (Proposition 23.1) and states its asymptotics as a question (item 1).

### 23.1 The third member of the family: A192562

Let  $c_n = \sum_{k=0}^n k! \begin{bmatrix} n \\ k \end{bmatrix} \{n+1\}_{k+1}$ , the sequence A192562 [18] (the write's letter;  $c_n$  is not used in this sense elsewhere in the report). Here  $a_n$ ,  $A_{n,k}$  and  $F_n$  are as in Part I.

**Proposition 23.1** ([write]). *For every  $n \geq 1$ ,*

$$c_n = a_n - nA_{n-1,n} = \sum_{m=0}^{n-1} \begin{bmatrix} n-1 \\ m \end{bmatrix} \{(m+2)^n - (m+1)^n\} = n! [z^n] (e^z - 1) F_{n-1}(z). \quad (23.1)$$

*In particular  $0 < c_n < a_n$ ; and  $c_0 = 1$ .*

*Proof.* All three equalities use the recurrence  $\begin{bmatrix} N+1 \\ i \end{bmatrix} = N \begin{bmatrix} N \\ i \end{bmatrix} + \begin{bmatrix} N \\ i-1 \end{bmatrix}$  ( $N \geq 0$ ,  $i \geq 1$ ;  $\begin{bmatrix} N \\ -1 \end{bmatrix} = 0$ ). *First equality.* Apply it with  $N = n$ ,  $i = k+1$  to the diagonal of (2.1):

$$a_n = \sum_{k=0}^n k! \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} \left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = n \sum_{k=0}^n k! \begin{bmatrix} n \\ k+1 \end{bmatrix} \left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} + \sum_{k=0}^n k! \begin{bmatrix} n \\ k \end{bmatrix} \left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\}.$$

The last sum is  $c_n$ . The first sum is (2.1) with  $(n, k)$  replaced by  $(n-1, n)$ , so it equals  $A_{n-1,n}$ . *Second equality.* Apply the recurrence with  $N = n-1$ ,  $i = m$  inside (1.1):

$$a_n = \sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix} (m+1)^n = (n-1) \sum_{m=0}^{n-1} \begin{bmatrix} n-1 \\ m \end{bmatrix} (m+1)^n + \sum_{m=0}^{n-1} \begin{bmatrix} n-1 \\ m \end{bmatrix} (m+2)^n,$$

and subtract  $nA_{n-1,n} = n \sum_m \begin{bmatrix} n-1 \\ m \end{bmatrix} (m+1)^n$ . *Third equality.* By (2.3),  $F_n(z) = (e^z + n-1)F_{n-1}(z)$ , so  $(e^z - 1)F_{n-1} = F_n - nF_{n-1}$ , and by (2.4)  $n! [z^n] F_n = A_{n,n} = a_n$  and  $n! [z^n] F_{n-1} = A_{n-1,n}$ . Finally, every term of the middle sum is nonnegative and the term  $m = n-1$  is  $(n+1)^n - n^n > 0$ ; and  $A_{n-1,n} \geq \begin{bmatrix} n-1 \\ n-1 \end{bmatrix} n^n > 0$ . For  $n = 0$  the definition gives  $c_0 = 0! \begin{bmatrix} 0 \\ 0 \end{bmatrix} \{1\} = 1$ .  $\square$

The write checked (23.1) by exact integers for  $1 \leq n \leq 55$ , and the 17 listed terms of A192562 against the definition. Since the first equality is a difference of two quantities that Part I's theorems describe, it is natural to ask how much cancellation it contains.

**Remark 23.2** ([write] A numerical observation). Write  $c_n = n! [z^n] F_n(z) \varphi_n(z)$  with  $\varphi_n(z) = (e^z - 1)/(e^z + n - 1)$ , and let  $r$  be Part I's saddle (1.4) at  $N = n$ ,  $t = e^r$ . By Lemma 3.3,  $t/n \rightarrow 0$ , so  $\varphi_n(r) \sim t/n \sim 1/(r \log r)$ . Exact  $c_n$  and  $a_n$ , with  $r$  solved to 40 digits (mpmath `findroot`), give:

| $n$ | $c_n/a_n$ | $\varphi_n(r)$ | $c_n/(a_n\varphi_n(r)) - 1$ | $(c_n/(a_n\varphi_n(r)) - 1) n/r$ |
|-----|-----------|----------------|-----------------------------|-----------------------------------|
| 10  | 0.261567  | 0.260095       | +0.0056605                  | +0.0375                           |
| 20  | 0.219065  | 0.218984       | +0.0003675                  | +0.0039                           |
| 50  | 0.171396  | 0.171640       | -0.0014187                  | -0.0292                           |
| 100 | 0.142211  | 0.142408       | -0.0013826                  | -0.0482                           |
| 200 | 0.118524  | 0.118649       | -0.0010546                  | -0.0633                           |
| 400 | 0.099549  | 0.099620       | -0.0007193                  | -0.0755                           |

At these six sizes the ratio  $c_n/a_n$  falls from 0.26 to 0.10, and  $a_n\varphi_n(r)$  matches  $c_n$  to a relative error below 0.006, which divided by  $r/n$  lies between  $-0.08$  and  $+0.04$ . These are floating-point observations at six sizes, not bounds; they suggest, but do not prove, the form stated in item 1 below.

## 23.2 The questions

1. **Asymptotics of A192562.** *Source:* the write; the intake recorded A192562 as treated by neither manuscript. *Question:* is

$$c_n = a_n \varphi_n(r) (1 + O(r/n)) = \frac{n! e^{L_n(r)} \varphi_n(r)}{r^n \sqrt{2\pi b}} (1 + O(r/n)),$$

with Part I's  $r$ ,  $b$ ,  $L_n$ , and is there an expansion to every fixed order? (The two forms agree by Theorem 1.1 at  $R = 1$ .) *Sketch, Part I's route:* by (23.1),  $c_n = n! [z^n] G_n(z)$  with  $G_n = (e^z - 1)F_{n-1}$ . On the circle  $|z| = r$ ,  $|e^z - 1| \leq e^{r \cos \theta} + 1$  and  $|F_{n-1}(re^{i\theta})| \leq F_{n-1}(r \cos \theta)$  by the positive exponential sum of  $F_{n-1}$ , so the argument of Lemma 4.2 bounds the complementary circle up to the factor  $(e^r + 1)/(e^r - 1)$ ; the exponential sum of  $G_n$  itself has coefficients of both signs, so Lemma 4.2 does not apply to it verbatim. On the disk  $|z - r| \leq \rho$  the factor  $e^z - 1$  is analytic and nonzero, with bounded logarithmic derivatives, so Lemma 4.1 extends to  $\log G_n$ . The central expansion then needs either the exact saddle of  $\log G_n(z) - n \log z$  or Part I's saddle with  $\varphi_n$  carried as an amplitude, whose Euler derivative  $(z\partial_z) \log \varphi_n$  is of order  $r$  at  $z = r$ . Either way the first correction relative to  $a_n\varphi_n(r)$  is expected to be of order  $r/n$ : the amplitude contributes terms such as  $((z\partial_z) \log \varphi_n)^2/b$  and  $(z\partial_z) \log \varphi_n \cdot \kappa_3/b^2$ , both  $O(r/n)$  by Lemma 3.3. *Missing:* the analogues of Lemmas 3.1–3.3 for the new phase (or the amplitude version of the weighted Taylor argument (4.10)), and the resulting coefficients. *Part II's route:*  $c_n = \langle z^{\bar{n}}, \mathfrak{T}_n \rangle_{\mathcal{F}}$  in the notation of “One family, two methods”, a bilinear pairing of two different polynomials, not a norm. The translation identity (13.6) extends to the pairing,  $\langle f, g \rangle_{\mathcal{F}} = e^{-b^2} f(b)g(b) \langle f_b, g_b \rangle_{\mathcal{F}}$  with  $f_b(t) = e^{-bt} f(b+t)/f(b)$ , but the terms  $k! [t^k] f_b [t^k] g_b$  need not be of one sign, so the positive certificate of Theorem 14.2 and its tail bound do not transfer;  $z^{\bar{n}}$  also vanishes at 0, outside Theorem 14.3. Cauchy–Schwarz gives only the upper bound  $c_n \leq \|z^{\bar{n}}\|_{\mathcal{F}} \|\mathfrak{T}_n\|_{\mathcal{F}}$ . *Missing:* a signed or two-centre version of the certificate. The same remarks apply to A192563 itself, which Part II's method does not treat, and conversely Part I's method has not been applied to A192561.

2. **Effective constants, onsets and certified thresholds.** *Sources:* Part I, question 1 and the last paragraph of Section 8; Part II, Subsection 22.3 and the statement after Theorem 19.1 that its constants and onset are not derived (“there is no claim that they are intrinsically noncomputable”). *Sketch:* Part I proposes making the disk, contour and weighted Taylor bounds explicit and combining them with interval evaluation of the gamma and polygamma functions; Part II's finite- $n$  certificate (Theorem 15.1 and the protocol of Subsection 19.4)

already decides any finite threshold for A192561 exactly. *Missing:* explicit constants and onsets for Part I's Theorems 1.1–1.3 and for Part II's (19.6) and Theorem 19.1, which would make the asymptotic windows usable at given  $X$  or  $T$ .

3. **Transition between moment regimes of A344639.** *Source:* Part I, question 2, and the closing paragraph of Section 6. *Question:* a uniform expansion of  $A_{n,k}$  as  $k/n \rightarrow 0$ , connecting fixed-order cycle moments with the growing-order regime, and separately as  $k/n \rightarrow \infty$  (fixed rows of the array). *Missing:* new uniform saddle and contour estimates; Part I warns that (1.11) may not be invoked there.
4. **Limit laws under the moment weights.** *Sources:* Part I, question 3; Part II, Subsection 22.4 (“weighted cycle-count distributions, local limits, and concentration”). *Question:* for the probability proportional to  $\binom{n}{m}(m+1)^k$ , a limit law and local estimates for the cycle count  $m$ ; for A192561, the law of the number  $k$  of matched pairs of cycles. *Missing:* everything; neither Part asserts a limit law.
5. **Further explicit inverse terms.** *Source:* Part I, question 4. *Sketch:* sharpen (9.3) by higher saddle-scale and gamma expansions and derive the nested-logarithm hierarchy with proved remainders, kept distinct from the much finer exact-saddle window. (Part II's centre  $v_R$  is already explicit at every fixed order.) *Missing:* the hierarchy and its remainders.
6. **Restricted and shifted families.** *Source:* Part I, question 5. *Sketch:* positivity may still control the global contour; the local derivative scales need separate proofs, after a full comparison with Yakymiv [9] and the shifted-family literature [7]. *Missing:* that comparison and the proofs.
7. **Growing orders, optimal truncation, exponentially small terms.** *Sources:* Part I, question 6 and the opening paragraph of Section 11; Part II, Subsection 22.3, and its statements that no convergence of the series in  $1/w$  and no growing-order bound is asserted (Section 12, after (12.7)) and that the existence of the coefficient generator is not a uniform estimate (after (18.12)). *Question:* the growth of Part I's contraction polynomials  $E_q$  and of Part II's  $A_r$  (Table 2 shows sign changes at  $r = 7, 9$ ), estimates uniform in the order, optimal truncation, exponentially small sectors, Borel summability. *Missing:* all of these; both Parts prove fixed orders only.
8. **The  $1/n$ -scale correction for A192561.** *Source:* Part II, Subsection 22.3. *Sketch:* Proposition 17.1 separates an  $O(w^2/n)$  finite-product correction from the expansion in  $1/w$ ; a systematic two-scale expansion would improve the forward formula and its inverse. The diagnostics in the note after Proposition 17.1 show this correction dominating the  $w^{-2}$  term at  $n \leq 1000$ . *Missing:* the first correction on the  $1/n$  scale and its coefficient.
9. **Root classes beyond the arithmetic progression.** *Source:* Part II, Subsection 22.1. *Question:* sharp uniform bounds on  $S_2$ , and decay rates, for regularly spaced roots, roots with a limiting density, or roots with a gap from zero; Theorem 14.3 gives only  $S_2 < 1$ , and the golden bound is specific to  $\lambda_j = j$ . Polynomials with nonreal or positive zeros need separate arguments. *Missing:* the bounds and those arguments.
10. **Complexity and interval algorithms.** *Source:* Part II, Subsection 22.2. *Question:* complexity in  $n$ ,  $K$  and the accuracy; ball arithmetic; range reduction for the exponential enclosures; adaptive centre and scale, in particular minimizing (15.3) over  $q$  instead of using

$K/(K+1)$ , which is admissible but not the minimizer off the saddle; a benchmark against the Stirling recurrence. *Missing*: all of these; the  $b_{10}$  example is not a speed claim.

11. **Other factorial weights.** *Source*: Part II, Subsection 22.4. *Question*:  $\sum_k \frac{[n+1]_k^2}{[k+1]} (k!)^\alpha$ , suggested by generalized Fock norms, which may need a different saddle geometry. *Missing*: everything beyond  $\alpha = 1$ .
12. **Literature and priority.** *Sources*: Part I, Subsection 2.3 and the close of Section 11; Part II, Section 20 and the remark after Theorem 14.3. *Missing*: full-text readings of Yakymiv [9] (moment-order uniformity), Komatsu–Szalay [7], Komatsu’s 2019 paper [8] and the final 2022 transform paper [5]; of Bender–Richmond [16] and Gao–Laferrière–Panario [17]; and a priority search for the formulation of Theorem 14.3. Both manuscripts state their searches as bounded; the write read none of these sources.

## References

- [1] OEIS Foundation Inc., *A192563*, official OEIS data mirror, entry last changed 26 May 2021, inspected 3 October 2026. [Official mirror entry](#). The inspected snapshot digest is recorded in the source provenance file. **[write]** The live entry (revision #13) and its b-file were read on 6 October 2026; see the front matter.
- [2] OEIS Foundation Inc., *A344639*, sign-normalized second-kind poly-Cauchy array, inspected 3 October 2026. [OEIS entry](#); [official mirror](#). **[write]** Revision #35, read on 6 October 2026.
- [3] B. Bényi and J. L. Ramírez, *Poly-Cauchy Numbers of the Second Kind—the Combinatorics Behind*, Enumerative Combinatorics and Applications **2**:1 (2022), Article S2R1, 13 pp. [DOI](#); [inspected final PDF](#). Final Theorems 2.1 and 2.6 are used here. The publisher records online publication in June 2021 for the 2022 volume.
- [4] T. Komatsu, *Poly-Cauchy numbers*, Kyushu Journal of Mathematics **67** (2013), 143–153. [DOI](#); [inspected final PDF](#).
- [5] T. Komatsu, *Recurrence relations of poly-Cauchy numbers by the  $r$ -Stirling transform*, arXiv:2103.15291 (2021), subsequently Mediterranean Journal of Mathematics **19** (2022). [Inspected preprint](#). A separate full final version was not inspected.
- [6] S. K. Jha, *Asymptotic expansions for factorial moments of some distributions in the analysis of algorithms*, arXiv:1611.07336 (2016). [Inspected preprint](#). The cycle-count moment order in Theorem 2.2 is fixed.
- [7] T. Komatsu and L. Szalay, *Shifted poly-Cauchy numbers*, Lithuanian Mathematical Journal **54**:2 (2014), 166–181. [DOI](#). Only repository and indexed extracts were inspected, including the bivariate formula associated with Proposition 3; full text uninspected.
- [8] T. Komatsu, *Some recurrence relations of poly-Cauchy numbers*, Journal of Nonlinear Sciences and Applications **12**:12 (2019), 829–845. [Publisher record](#). Abstract and indexed theorem only; full text uninspected.
- [9] A. L. Yakymiv, *Asymptotics with remainder term for moments of the total cycle number of random  $A$ -permutation*, Diskretnaya Matematika **31**:3 (2019), 114–127; English translation,

- Discrete Mathematics and Applications **31**:1 (2021), 51–60. [Russian DOI](#); [translation DOI](#). Official abstract inspected; full text and uniformity in moment order unverified.
- [10] NIST Digital Library of Mathematical Functions, Chapter 5, *Gamma Function*, equations [5.9.13](#), [5.9.16](#), [5.11.1](#), and [Section 5.11\(ii\)](#); verified 3 October 2026. The fixed-derivative bound used here is derived directly from the integral remainder. Standard fuller expansions also appear in [5.11.2](#) and [5.15.9](#).
  - [11] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A192561, accessed 3 October 2026. <https://oeis.org/A192561>. Official data mirror: <https://github.com/oeis/oeisdata/blob/main/seq/A192/A192561.seq>. [write] Revision #10, read on 6 October 2026.
  - [12] H. S. Wilf, *The variance of the Stirling cycle numbers*, arXiv:math/0511428, 2005. <https://arxiv.org/abs/math/0511428>.
  - [13] D. Alpay and M. Porat, *Generalized Fock spaces and the Stirling numbers*, arXiv:1804.05254, 2018. <https://arxiv.org/abs/1804.05254>.
  - [14] NIST Digital Library of Mathematical Functions, Sections 5.11 and 25.11, *Asymptotic Expansions*, especially Eqs. 5.11.1, 5.11.2, and 5.11.8. <https://dlmf.nist.gov/5.11>; Hurwitz-zeta expansions: <https://dlmf.nist.gov/25.11>.
  - [15] B. Gittenberger and J. Mandlbürger, *Hayman admissible functions in several variables*, author preprint, 2006. <https://dmg.tuwien.ac.at/bgitten/preprints/multi-H.pdf>.
  - [16] E. A. Bender and L. B. Richmond, *Admissible Functions and Asymptotics for Labelled Structures by Number of Components*, *Electronic Journal of Combinatorics* **3**(1), R34, 1996. <https://doi.org/10.37236/1258>. Full-text inspection incomplete; see Section 20.
  - [17] Z. Gao, D. Laferrière, and D. Panario, *Asymptotics of Decomposable Combinatorial Structures of Alg-Log Type With Positive Log Exponent*, 2010. <https://doi.org/10.46298/dmtcs.2798>. Full-text inspection incomplete; see Section 20.
  - [18] [write] OEIS Foundation Inc., A192562. <https://oeis.org/A192562>. Revision #4 of 30 March 2012, read on 6 October 2026.
  - [19] [write] *Transseries and inversion*, ProveIt volume, Analysis/Transseries/docs/series-and-transseries/Transseries\_And\_Inversion/transseries\_and\_inversion.tex: Definitions p0:def:model and plt:def:lw-monomial-log-datum, Theorems p0:thm:lambert-core, p0:thm:staircase and plt:thm:lw-template, and Proposition p0:prop:factorial-core.